Portugal's Economic and Human Development: A Century of Transformation

Data Analysis and Integration · 2026/2027

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Task 1 – Staging Database and Data Profiling

Approach

- An isolated staging schema, `DevelopmentDB`, was implemented as a landing layer between the CSV files and the warehouse. Data was loaded as-is. One table per source, so each load can be reconciled with its file. Constraints (keys, types, NOT NULL) reject invalid rows at load instead of staging them.
- Profiling queries check the load, integrity and coverage. Their findings feed the data dictionaries and the design decisions that follow.

Design Choices and Justification

Element	Choice	Justification
Primary key, indicator tables	`(country_iso3 indicator_code, year)` (composite)	Unique in the data, so no surrogate key is needed. Duplicate rows in a source file are rejected at load instead of staged.
Primary key, events	Surrogate `event_id AUTO_INCREMENT`	`(country_iso3, year)` is not unique: 24 country-years have several events. Adding `event` makes it unique, but it is a text column, too long for a key.
`country_iso3`	`CHAR(3)`	Codes are always 3 letters. Fixed length fits the data; exceeding 3 characters is rejected at load. Code validity is checked in profiling against the expected country list.
`value`	`DECIMAL(24,6)`	Fixed-point storage avoids floating-point precision issues during numerical aggregation. The precision accommodates the largest observed indicator.
`year`	`INT`	The source stores calendar years as four-digit integer values. INT represents this format directly and supports numerical operations.
Text columns	`VARCHAR`, sized appropriate to the longest observed value	Variable-length fields reduce unnecessary storage while permitting the observed descriptions, labels, and event text.
`NOT NULL`	All source columns	Enforces validity at load: a row with a missing value is rejected instead of staged.
`loaded_at`	`TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP`	Records when the current content was staged, so a warehouse value can be traced to the load that produced it.

Relationship Between the Sources

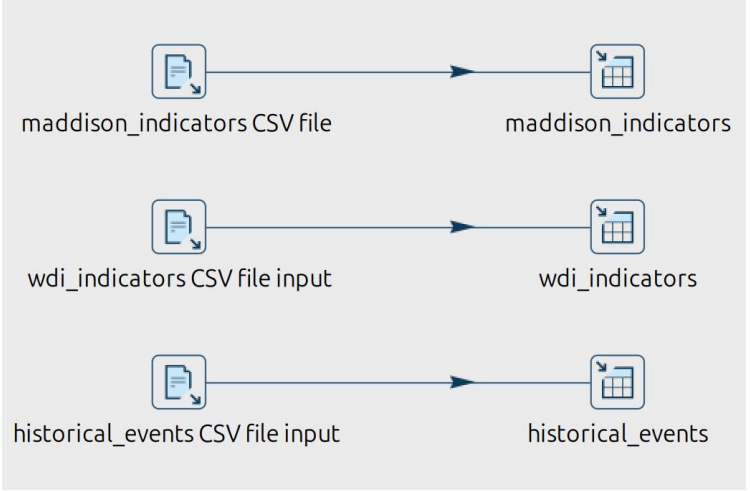
Characteristic	maddison_indicators	wdi_indicators	historical_events
Number of records	4,349	11,371	386*
Number of columns	8	8	8
Nature	Long-run historical estimates	Annual statistical indicators	Dated qualitative events
Period	1900 – 2022	1960 – 2025	1900 – 2025
Entities	15 countries	15 countries + EUU	15 countries + EUU
Unit of observation	Country × indicator × year	Country × indicator × year	One event; up to 5 per country-year
Measures	GDP, GDP per capita, population	Growth, trade, GNI per capita, life expectancy, urbanisation, sector value added and employment	Event text, category, economic impact
Portugal	Full series, 1900 – 2022	From 1960 (sector value added from 1995)	125 events (32% of all)
Role in the warehouse	FACT_ECONOMY, FACT_SOCIETY	FACT_ECONOMY, FACT_SOCIETY, FACT_SECTOR	Context linked by country and year (approach chosen in Task 3)

*The initial file contained 387 records, however, one of them was malformed and, therefore, rejected.

Load Settings (Apache Hop)

- At first, a database connection to `DevelopmentDB` was created.
- The initial Apache Hop pipeline (before exploratory queries) contained three ingestion flows, one for each source dataset. Each flow read its CSV file, applied column selection and type alignment, and wrote records to the relevant staging table.

Transform	Setting	Value	Why
CSV file input	Delimiter / enclosure	,	File format.
CSV file input	Header row present	Yes	The first line holds column names, not data
CSV file input	Encoding	UTF-8	38 events contain accented characters
CSV file input	Field types	year = Integer, value = BigNumber, other fields = String	Match the staging columns; BigNumber keeps DECIMAL values exact
Table output	Connection / target table	DevelopmentDB / one table per file	One flow per source, so each load is reconciled with its file
Table output	Database fields	CSV fields mapped by name; loaded_at not mapped	The database fills loaded_at (DEFAULT CURRENT_TIMESTAMP)



Hop pipeline (initial approach – before exploratory queries)

Exploratory Queries

- After staging, exploratory queries were run to **validate** the process, **explore** the datasets to **inform** the data dictionaries, and **identify** data inconsistencies or invalid records.
- Exploratory queries were used to assess each staged dataset by checking data volume, country and indicator coverage, and year ranges. Additional checks identified missing values, null fields, negative values, duplicate records and inconsistencies in historical event data. The results also helped understand dataset characteristics and define the content of the data dictionaries.

```
SELECT country_iso3, country_name, COUNT(*) AS total_rows,
       MIN(year) AS first_year, MAX(year) AS last_year
FROM maddison_indicators
GROUP BY country_iso3, country_name
ORDER BY country_iso3;



SELECT indicator_code, COUNT(*) AS total_rows, COUNT(DISTINCT
country_iso3) AS entities,
       MIN(year) AS first_year, MAX(year) AS last_year
FROM maddison_indicators
GROUP BY indicator_code;
```

country_iso3	country_name	total_rows	first_year	last_year
BEL	Belgium	369	1900	2022
CZE	Czechia	159	1970	2022
DEU	Germany	369	1900	2022
ESP	Spain	369	1900	2022
EST	Estonia	132	1973	2022
FRA	France	369	1900	2022
GRC	Greece	369	1900	2022
HUN	Hungary	300	1900	2022
IRL	Ireland	309	1913	2022
ITA	Italy	369	1900	2022
LTU	Lithuania	132	1973	2022
LVA	Latvia	132	1973	2022
POL	Poland	267	1900	2022
PRT	Portugal	369	1900	2022
ROU	Romania	335	1900	2022

indicator_code	total_rows	entities	first_year	last_year
gdp	1444	15	1900	2022
gdppc	1461	15	1900	2022
pop	1444	15	1900	2022

Exploratory Queries

```
SELECT category, COUNT(*) AS total_events
FROM historical_events
GROUP BY category
ORDER BY category;


SELECT 'maddison' AS source, country_iso3,
indicator_code, MIN(year) AS first_year,
       MAX(year) AS last_year, MAX(year) - MIN(year) + 1
- COUNT(*) AS missing_years
FROM maddison_indicators
GROUP BY country_iso3, indicator_code
HAVING MAX(year) - MIN(year) + 1 <> COUNT(*)
UNION ALL
SELECT 'wdi', country_iso3, indicator_code, MIN(year),
MAX(year),
       MAX(year) - MIN(year) + 1 - COUNT(*)
FROM wdi_indicators
GROUP BY country_iso3, indicator_code
HAVING MAX(year) - MIN(year) + 1 <> COUNT(*);
```

category	total_events
Colonial	5
Economy	90
Financial	32
Geopolitical	13
Industry	20
Infrastructure	31
Infrastructure/Economy	1
Integration	40
Integration/Industry	1
Integration/Security	1
Political	102
Social	10
War	39
War/Health	1
War/Industry	1

source	country_iso3	indicator_code	first_year	last_year	missing_years
maddison	EST	gdp	1973	2022	6
maddison	EST	gdppc	1973	2022	6
maddison	EST	pop	1973	2022	6
maddison	HUN	gdp	1900	2022	23
maddison	HUN	gdppc	1900	2022	23
maddison	HUN	pop	1900	2022	23
maddison	IRL	gdp	1913	2022	7
maddison	IRL	gdppc	1913	2022	7
maddison	IRL	pop	1913	2022	7
maddison	LTU	gdp	1973	2022	6
maddison	LTU	gdppc	1973	2022	6
maddison	LTU	pop	1973	2022	6
maddison	LVA	gdp	1973	2022	6
maddison	LVA	gdppc	1973	2022	6
maddison	LVA	pop	1973	2022	6
maddison	POL	gdp	1900	2022	34
maddison	POL	gdppc	1900	2022	34
maddison	POL	pop	1900	2022	34
maddison	ROU	gdp	1900	2022	17
maddison	ROU	pop	1900	2022	17

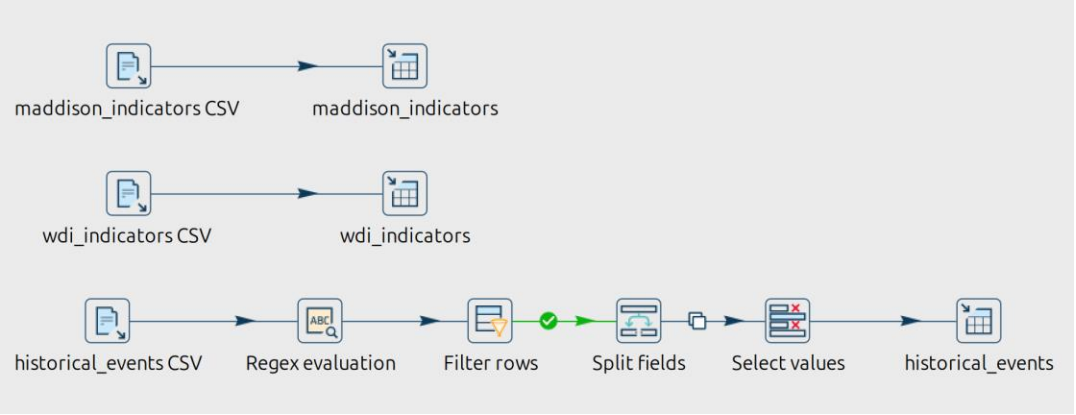
Source Defect and corrective approach

- When performing exploratory queries, one invalidly structured source record was identified in the `historical_events` file for Ireland, in 1990. The record contains a comma in the event description (record bellow).

```
1990,IRL,Ireland,Celtic Tiger boom begins; strong FDI, IT and pharmaceuticals growth through the 1990s,Economy,High growth
```

- The comma after FDI was interpreted it as a field delimiter. Consequently, the record was read as 7 fields rather than the expected 6 . The CSV load does not fail: it splits on every comma, shifts the columns and drops the last field, and the row count stays correct (387).

- To address this issue, the Apache Hop pipeline was modified to validate records before loading them into the staging table. Our approach was to **reject** (read ! Note) that record:
  1. `Text File Input`: Reads each source line as a single field (`line`), preserving the original record before parsing.
  2. `Regex Evaluation`: Checks whether each line contains exactly 5 comma-separated fields, generating a `result` flag.
  3. `Filter Rows`: Routes records according to the validation result. Records marked as valid (Y) proceed to the loading process, while invalid records (N) are excluded from it.
  4. `Split Fields`: Splits each valid line into the six expected fields: `year`, `country_iso3`, `country_name`, `event`, `category`, and `economic_impact`.
  5. `Select Values`: Retains only the six fields required by the staging table (removing the intermediate `result` field).
  6. `Table Output`: Loads the validated and correctly structured records into the staging table.



Hop pipeline (updated)

! Note: *In a real-world scenario, invalid records should be routed to a dedicated rejection file or error table for inspection, correction, and potential reprocessing. For the purposes of this project, the invalid record is simply excluded from the load, as the objective is to ensure that only correctly structured records enter the staging table.*

Detecting the Defect (query and output)

```
SELECT category, COUNT(*) AS total_events
FROM historical_events
GROUP BY category
ORDER BY category;
```

Before the correction: 16 categories

category	total_events
IT and pharmaceuticals growth through the 1990s	**1**
Colonial	5
Economy	89
Financial	32
Geopolitical	13
Industry	20
Infrastructure	31
Infrastructure/Economy	1
Integration	40
Integration/Industry	1
Integration/Security	1
Political	102
Social	10
War	39
War/Health	1
War/Industry	1

The faulty record appears as a category that is a fragment of the event text (its leading space sorts it first).

After the correction: 15 categories

category	total_events
Colonial	5
Economy	89
Financial	32
Geopolitical	13
Industry	20
Infrastructure	31
Infrastructure/Economy	1
Integration	40
Integration/Industry	1
Integration/Security	1
Political	102
Social	10
War	39
War/Health	1
War/Industry	1

As 1 record was rejected, total counts was updated to 386 records.

Data Profiling Insights (& data dictionaries info)

- After running exploratory queries, which informed the development of the data dictionaries, several relevant findings were identified. Beyond statistical summaries, the queries revealed specific data irregularities and coverage limitations. The most relevant findings for the subsequent analysis are presented below.

Maddison Project	World Development Indicators (WDI)	Historical Events
EU aggregate missing: EUU exists in WDI and historical events but not in Maddison.	**Unequal sectoral value-added coverage**: France starts in 1960; Italy and Romania in 1990; Germany and EUU in 1991; Czechia in 1993; remaining entities in 1995.	**Portugal-focused distribution**: Portugal accounts for 125 of 386 event records, reflecting the project's focus.
GDP per capita without corresponding GDP: Romania has 17 observations without GDP values, spanning 1901–1919.	**Employment data start late**: Agriculture, industry, and services employment cover 1991–2025 only.	**Events predate WDI**: 165 events occurred before 1960, when WDI coverage begins.
Uneven temporal coverage: GDP per capita starts in 1913 for Ireland, 1970 for Czechia, and 1973 for Estonia, Latvia, and Lithuania.	**Manufacturing employment unavailable**: No WDI indicator code matches manufacturing employment.	**Partial temporal overlap**: Earlier events can be compared with Maddison only where matching country-year observations exist.
Internal gaps: Estonia, Hungary, Ireland, Latvia, Lithuania, Poland, and Romania have missing years.	**GDP growth includes valid negative values**: The range is −32.1% to 42.4%; negative values represent contractions, not necessarily errors.	
Substantial gaps in Hungary and Poland: Hungary has 23 missing years per indicator; Poland has 34.	**Incomplete temporal coverage**: Trade and sectoral value-added series have gaps, particularly in earlier years.	

Task 2 – Country-Code Reconciliation

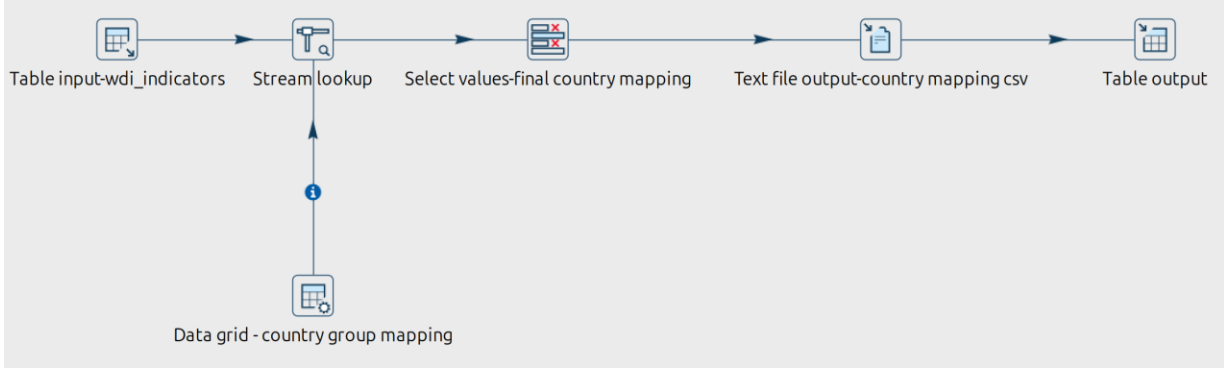
Approach

- The sources already share ISO3 codes and names , so one source is enough if it contains every entity. WDI does: all 16 (Maddison has 15, without EUU), so we used it as input.
- We used `Data grid` to define country groups as defined in the project statement, joined with a `stream lookup;`
- The staging table on `DevelopmentDB` was created form Hop's generated SQL (`Table output` -> SQL)

Pipeline steps

- The country_matching.hpl consist in:

1. `Table input` to read the distinct countries from the staging table of WDI. The configuration was:
    - Connection: `DevelopmentDB`
    - SQL: `SELECT DISTINCT country_iso3, country_name FROM wdi_indicators`
2. `Data grid` to map each country code to its group. In the Meta tab, two fields were given: `iso3_code` and `country_group` (both String). Then the Data tab followed the configuration according to the project assignment.
3. `Stream lookup` to add the group to each country. The Table input is the main stream and the Data grid is the lookup stream.
4. `Select values` to rename and order the fields.
5. `Text file output` to write the mapping to a CSV file. `Table output` to load the same mapping into the staging database.



Hop pipeline

Task 3 – Data Warehouse Design

Schema Overview and Design Principals

Table	Type	Content [Grain]
DIM_COUNTRY	Dimension	15 countries + EUU, with country group (Task 2 mapping)
DIM_TIME	Dimension	Years 1900–2025, with decade
DIM_SECTOR	Dimension	Agriculture, Industry, Manufacturing, Services, with broad sector
FACT_ECONOMY	Fact	GDP and GDP per capita (Maddison); GDP growth, exports, imports (WDI) [country x year]
FACT_SOCIETY	Fact	Population (Maddison); GNI per capita, life expectancy, urbanisation (WDI) [country x year]
FACT_SECTOR	Fact	Value added and employment by sector (WDI) [country x year x sector]
BRIDGE_EVENT	Bridge	Historical events, linked to DIM_COUNTRY and DIM_TIME

- Country and time are shared by every fact table, so the subject areas can be compared by country and year.
- Surrogate key on every dimension, as required. Natural keys (`ISO3_code`, `year`, `sector_name`, `event_id`) are enforced as UNIQUE, preventing duplicate dimension members and supporting unambiguous lookups during ETL loading.
- Each fact table's primary key is its grain (the combination of its dimension keys): a duplicate fact row is rejected.
- Dimension attributes are `NOT NULL`; fact measures are nullable. A missing observation stays empty and is never replaced by 0.
- Foreign keys enforce referential integrity.

Design Decisions

Detailed justification on design_justification.pdf

Design decision	Justification
Three fact tables	Separate analytical subjects and preserve their natural grains: country-year for `FACT_ECONOMY` and `FACT_SOCIETY`, and country-year-sector for `FACT_SECTOR`.
Conformed dimensions	`DIM_COUNTRY` and `DIM_TIME` are shared across facts, enabling consistent geographical and temporal analysis.
`DIM_SECTOR` only for `FACT_SECTOR`	Sector is part of the grain of `FACT_SECTOR`; the other fact tables contain only country-year measures.
Historical events — Option 2	`BRIDGE_EVENT` preserves individual event records while linking them to `DIM_COUNTRY` and `DIM_TIME`. This provides temporal and geographical integration without changing the grain of the quantitative facts.
Surrogate keys	Every dimension uses a surrogate key, providing warehouse-specific identifiers independent of source-system keys.
Unique natural keys	`iso3_code`, `year`, `event_id` and `sector_name` are enforced as `UNIQUE` to prevent duplicate dimension members and support unambiguous ETL lookups.
Nullable measures	Fact measures may be `NULL` to preserve genuine coverage gaps and is not interpreted as zero.
Mandatory dimensional keys	Country, time and sector foreign keys are required for every fact row, ensuring that each observation has an unambiguous dimensional context.
Data types	Numeric measures use `DECIMAL` to preserve precision; keys and years use integer types; codes and descriptive attributes use character types appropriate to their expected values.
No unnecessary source attributes	The warehouse retains attributes and measures required for the defined analytical model; source-specific metadata and raw fields remain in the staging layer.

Task 4 – ETL Pipelines

DIM_COUNTRY

- At first, a new database connection was created to `DevelopmentDW` data warehouse.
- Pipeline dim_coutry.hpl consists in:
  1. `Table input` to read the `country_mapping` table, and retrieve the following fields: `iso3_code, coutry_name and country_goup;`
  2. `Insert / update` to populate the country dimension table in data warehouse. The configuration was:
     - Connection: `DevelopmentDW`
     - Table target: `DIM_COUNTRY`
     - The key(s) to the look up value(s):

Table field	Comparator	Stream field 1
iso3_code	=	iso3_code

     - Update fields:

Table field	Stream field	Update
iso3_code	iso3_code	Y
country_name	country_name	Y
country_group	country_group	Y



Hop pipeline

DIM_TIME

- Pipeline dim_time.hpl consist in:

1. `Table Input` to read the `year` field from the `maddison_indicators`, `wdi_indicators`, and `historical_events` tables, and retrieve the minimum and maximum year across all three sources;
2. `Calculator` to compute number of year between max and min year;
3. `Clone rows` to replicate number of years in rows (clone num field: `offset`);
4. Another `calculator` to compute `decade`, using the following:

New Field	Calculation	Field A	Field B	Field C	Value Type
year	A + B	min_year	offset	–	Integer
ten	Set field to constant value A	10	–	–	Integer
year_div	A / B	year	ten	–	Number
decade_floor	FLOOR(A)	year_div	–	–	Number
decade	A * B	decade_floor	ten	–	Integer

5. `Insert / update` to populate the country dimension table in data warehouse. The configuration was:
   - Connection: `DevelopmentDW;` Table target: `DIM_TIME`
   - The key(s) to the look up value(s):

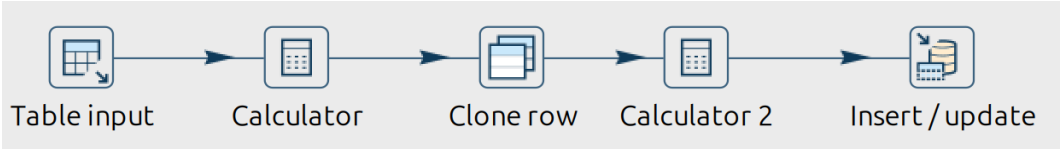
Table field	Comparator	Stream field 1
year	=	year

   - Update fields:

Table field	Stream field	Update
year	year	Y
decade	decade	Y

```
SELECT MIN(year) AS min_year,
       MAX(year) AS max_year
FROM (
    SELECT year FROM maddison_indicators
    UNION ALL
    SELECT year FROM wdi_indicators
    UNION ALL
    SELECT year FROM historical_events
) AS all_years;
```

`Table Input` query



Hop pipeline

DIM_SECTOR

- Pipeline dim_sector.hpl consists in:

1. `Data grid` to map `sector_name` with `broad_sector`. In Meta 2 those two field names were given. Then, Data followed the configuration:

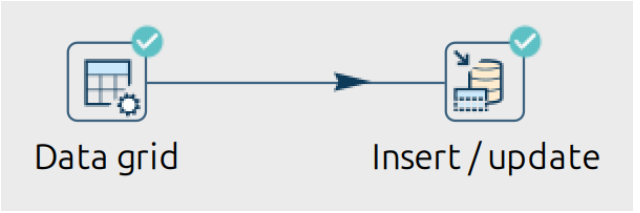
sector_name	broad_sector
Agriculture	Primary
Industry	Secondary
Manufacturing	Secondary
Services	Tertiary

2. `Insert / update` to populate the sector dimension table in data warehouse. The configuration was:
   - Connection: `DevelopmentDW`
   - Table target: `DIM_SECTOR`
   - The key(s) to the look up value(s):

Table field	Comparator	Stream field 1
sector_name	=	sector_name

   - Update fields:

Table field	Stream field	Update
sector_name	sector_name	Y
broad_sector	broad_sector	Y



Hop pipeline

Fact Tables – indicators' mapping

FACT_SOCIETY

Staging table	indicator_code	Measure
maddison_indicators	pop	population
wdi_indicators	NY.GNP.PCAP.CD	gni_per_capita
wdi_indicators	SP.DYN.LE00.IN	life_expectancy
wdi_indicators	SP.URB.TOTL.IN.ZS	urban_pop_pct

FACT_ECONOMY

Staging table	indicator_code	Measure
maddison_indicators	gdp	gdp
maddison_indicators	gdppc	gdp_per_capita
wdi_indicators	NY.GDP.MKTP.KD.ZG	gdp_growth_pct
wdi_indicators	NE.EXP.GNFS.ZS	exports_pct_gdp
wdi_indicators	NE.IMP.GNFS.ZS	imports_pct_gdp

FACT_SECTOR

Staging table	indicator_code	Sector	Measure
wdi_indicators	NV.AGR.TOTL.CD	Agriculture	value_added_usd
wdi_indicators	NV.AGR.TOTL.ZS	Agriculture	value_added_pct_gdp
wdi_indicators	SL.AGR.EMPL.ZS	Agriculture	employment_pct
wdi_indicators	NV.IND.TOTL.CD	Industry	value_added_usd
wdi_indicators	NV.IND.TOTL.ZS	Industry	value_added_pct_gdp
wdi_indicators	SL.IND.EMPL.ZS	Industry	employment_pct
wdi_indicators	NV.IND.MANF.CD	Manufacturing	value_added_usd
wdi_indicators	NV.IND.MANF.ZS	Manufacturing	value_added_pct_gdp
wdi_indicators	NV.SRV.TOTL.CD	Services	value_added_usd
wdi_indicators	NV.SRV.TOTL.ZS	Services	value_added_pct_gdp
wdi_indicators	SL.SRV.EMPL.ZS	Services	employment_pct

FACT_ECONOMY & FACT_SOCIETY

- FACT_ECONOMY and FACT_SOCIETY follow the same chain with adapted parameters for each;
  1. `Table Input` returns the data in long format: `country_iso3, year, indicator_code` (the ones relevant depending on the fact table), `value`

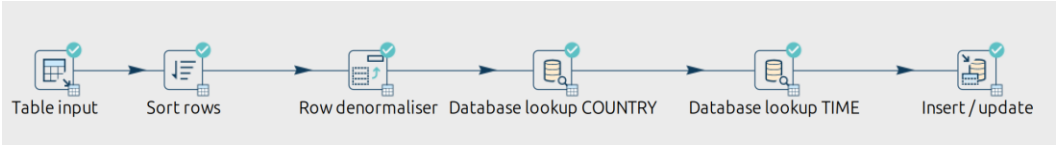
FACT_ECONOMY

```
SELECT country_iso3, year, indicator_code, value
FROM maddison_indicators
WHERE indicator_code IN ('gdppc', 'gdp')
UNION ALL
SELECT country_iso3, year, indicator_code, value
FROM wdi_indicators
WHERE indicator_code IN ('NY.GDP.MKTP.KD.ZG', 'NE.EXP.GNFS.ZS',
'NE.IMP.GNFS.ZS')
```

FACT_SOCIETY

```
SELECT country_iso3, year, indicator_code, value
FROM maddison_indicators
WHERE indicator_code = 'pop'
UNION ALL
SELECT country_iso3, year, indicator_code, value
FROM wdi_indicators
WHERE indicator_code IN ('NY.GNP.PCAP.CD', 'SP.DYN.LE00.IN',
'SP.URB.TOTL.IN.ZS')
```

  2. `Sort rows` by `country_iso3, year` (required by row denormaliser)
  3. `Row denormaliser` pivots the rows into columns:
     - Group fields: `country_iso3, year;`
     - Key field: `indicator_code`
     - One target field per measure (slide "Fact Tables – indicators' mapping")



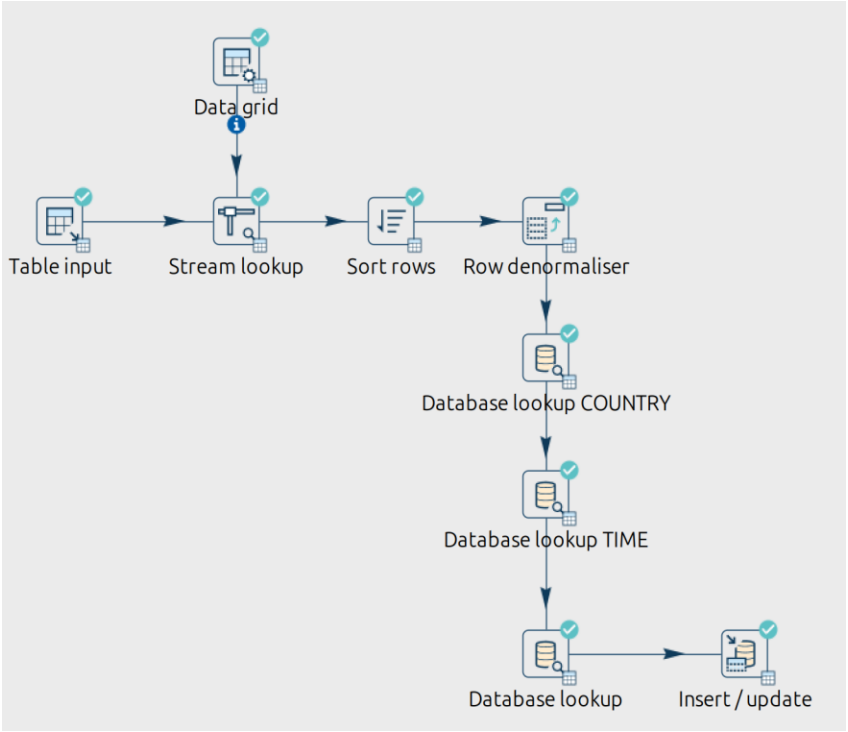
Hop pipeline

  4. `Database lookup` (`DevelopmentDW`), one for each dimension:
     - Country: table `DIM_COUNTRY`, key `iso3_code = country_iso3`, returns `country_key;`
     - Time: table `DIM_TIME`, key `year = year`, returns `time_key`
  5. `Update/ Insert`:
     - Connections to `DevelopmentDW`; Target table `FACT_X` (X = ECONOMY or SOCIETY)
     - Keys to lookup: `country_key = country_key` and `time_key = time_key`;
     - Update fields: indicators, setting update to Y.

FACT_SECTOR

- FACT_SECTOR pipeline comprises the following steps;

1. `Table Input` returns the data in long format: `country_iso3, year, indicator_code` (the ones relevant ones for the fact table), `value;`
2. `Data Grid,` with 3 string fields `indicator_code, sector_name` and `measure` with 11 rows (slide "Fact Tables – indicators' mapping")
3. `Stream lookup:`
   - `Table input` is connected as the main stream, and the Data grid as the lookup step.
   - Keys: Field `indicator_code`, Lookup field `indicator_code`.
   - Values to retrieve: `sector_name` and `measure`, both String.
4. `Sort rows` by `country_iso3, year, sector_name;`
5. `Row denormaliser` pivots the rows into columns:
   - Group fields: `country_iso3, year, sector_name;`
   - Key field: measure
   - One target field per measure (slide "Fact Tables – indicators' mapping")
6. `Database lookup` (`DevelopmentDW`), one for each dimension:
   - Country: table `DIM_COUNTRY,` key `iso3_code = country_iso3,` returns `country_key;`
   - Time: table `DIM_TIME,` key `year = year,` returns `time_key`
   - Sector: table `DIM_SECTOR,` key `sector_name = sector_name,` returns `sector_key;`
7. `Update/ Insert,` connections to `DevelopmentDW`; Target table `FACT_SECTOR`
   - Keys to lookup: `country_key = country_key, time_key = time_key, sector_key = sector_key;`
   - Update fields: indicators, setting update to Y.

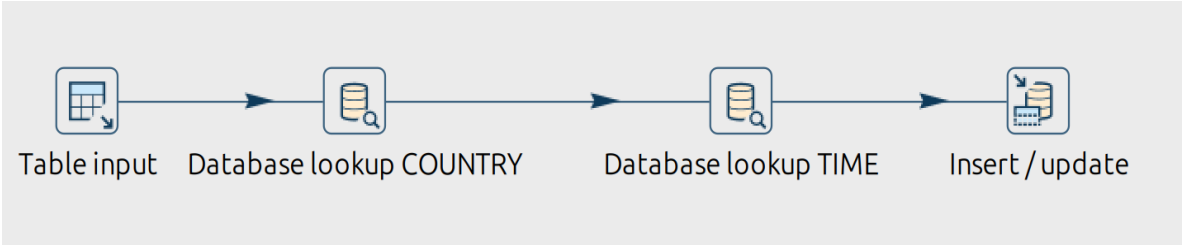


Hop pipeline

EVENTS

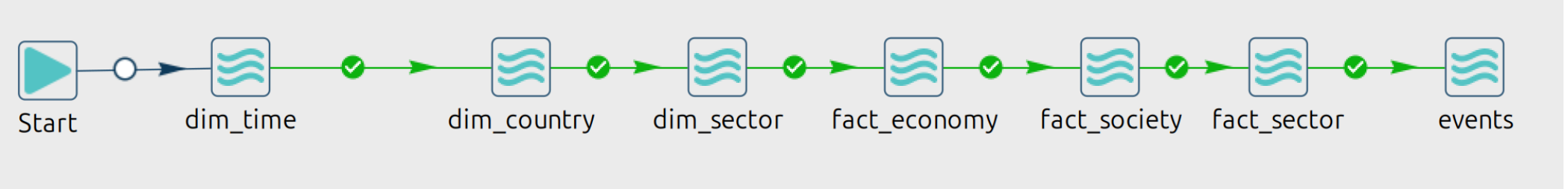
- Events pipeline comprises the following steps;
  1. `Table Input` read the historical_events table data in long format: `country_iso3, event_id, year, event, category, economic_impact;`
  2. `Database lookup` (`DevelopmentDW`), one for each dimension:
     - Country: table `DIM_COUNTRY`, key `iso3_code = country_iso3`, returns `country_key;`
     - Time: table `DIM_TIME`, key `year = year`, returns `time_key`
  3. `Update/ Insert, c`onnections to `DevelopmentDW`; Target table BRIDGE_EVENTS
     - Keys to lookup: `event_id = event_id;`
     - Update fields:

Table field	Stream field	Update
event_id	event_id	N
country_key	country_key	Y
time_key	time_key	Y
event	event	Y
category	category	Y
economic_impact	economic_impact	Y



Hop pipeline

Workflow



Validation

- Record counts were compared between the staging data and the populated data warehouse.
- Dimension tables were validated against their expected members.
- For the three fact tables, the approach was to select the indicators that supply the measures required by the warehouse (slide "Fact Tables – indicators' mapping") , identify the distinct combinations matching each table's grain, and count those combinations.

Table (Warehouse)	# Rows	Justification of row count
DIM_COUNTRY	16	One record per country / entity
DIM_TIME	126	One record per year from 1900 to 2025, inclusive
DIM_SECTOR	4	One record for each sector: Agriculture, Industry, Manufacturing and Services
FACT_ECONOMY	1,571	Select the relevant Maddison indicators and WDI indicators (slides' section 4.4). Combine their country-year combinations using UNION, which removes duplicates, and count the resulting rows
FACT_SOCIETY	1,622	
FACT_SECTOR	2,329	Select relevant WDI indicators. Map the indicators to their corresponding sectors, select distinct country-year-sector combinations, and count them. Multiple indicators for the same sector and country-year contribute to one fact row
BRIDGE_EVENT	386	One record per historical event in the source dataset (nª records = nº of records in historical_events)

Validation (queries)

Staging, `DevelopmentDB`**, record counts**

```
SELECT COUNT(*) AS expect_fsociety
FROM (
  SELECT country_iso3, year
  FROM maddison_indicators
  WHERE indicator_code = 'pop'
  UNION
  SELECT country_iso3, year
  FROM wdi_indicators
  WHERE indicator_code IN (
    'NY.GNP.PCAP.CD',
    'SP.DYN.LE00.IN',
    'SP.URB.TOTL.IN.ZS'
  )
  AND value IS NOT NULL ) AS society;
```

FACT_SOCIETY

```
SELECT COUNT(*) AS expect_feconomy
FROM (
  SELECT country_iso3, year
  FROM maddison_indicators
  WHERE indicator_code IN ('gdp', 'gdpp')
  UNION
  SELECT country_iso3, year
  FROM wdi_indicators
  WHERE indicator_code IN (
    'NY.GNP.MKTP.KD.ZG',
    'NE.EXP.GNFS.ZS',
    'NE.IMP.GNFS.ZS'
  )
  AND value IS NOT NULL ) AS economy;
```

FACT_ECONOMY

```
SELECT COUNT(*) AS expect_fsector
FROM ( SELECT DISTINCT country_iso3, year,
    CASE WHEN indicator_code LIKE 'NV.AGR.%'
       OR indicator_code = 'SL.AGR.EMPL.ZS' THEN
'Agriculture'
     WHEN indicator_code LIKE 'NV.IND.MANF.%' THEN
'Manufacturing'
     WHEN indicator_code LIKE 'NV.IND.%'
       OR indicator_code = 'SL.IND.EMPL.ZS' THEN
'Industry'
     ELSE 'Services' END AS sector
  FROM wdi_indicators
  WHERE indicator_code IN (
    'NV.AGR.TOTL.CD', 'NV.AGR.TOTL.ZS', 'SL.AGR.EMPL.ZS',
    'NV.IND.TOTL.CD', 'NV.IND.TOTL.ZS', 'SL.IND.EMPL.ZS',
    'NV.IND.MANF.CD', 'NV.IND.MANF.ZS', 'NV.SRV.TOTL.CD',
    'NV.SRV.TOTL.ZS', 'SL.SRV.EMPL.ZS' )
  AND value IS NOT NULL
) AS sectors;
```

FACT_SECTOR

Warehouse, `DevelopmentDW`**, record counts**

```
SELECT 'DIM_COUNTRY' AS table_name, COUNT(*) AS records
FROM DIM_COUNTRY
UNION ALL
SELECT 'DIM_TIME', COUNT(*) FROM DIM_TIME
UNION ALL
SELECT 'DIM_SECTOR', COUNT(*) FROM DIM_SECTOR
UNION ALL
SELECT 'FACT_ECONOMY', COUNT(*) FROM FACT_ECONOMY
UNION ALL
SELECT 'FACT_SOCIETY', COUNT(*) FROM FACT_SOCIETY
UNION ALL
SELECT 'FACT_SECTOR', COUNT(*) FROM FACT_SECTOR
UNION ALL
SELECT 'BRIDGE_EVENT', COUNT(*) FROM BRIDGE_EVENT;
```

- The outputs of the staging database queries (above) were compared with the ones from the warehouse (left).

Validation conclusion:

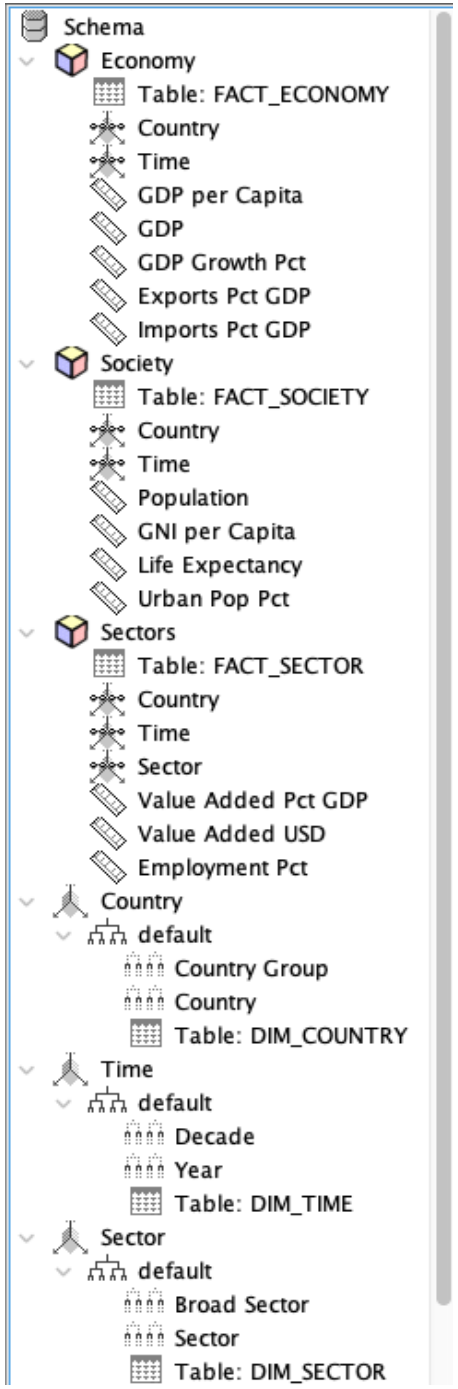
The row counts match the expected records across all warehouse tables: `DIM_COUNTRY` (16), `DIM_TIME` (126), `DIM_SECTOR` (4), `FACT_ECONOMY` (1,571), `FACT_SOCIETY` (1,622), `FACT_SECTOR` (2,329), and `BRIDGE_EVENT` (386). This **confirms consistency** between the expected dimensional members, selected source indicator combinations, and loaded warehouse records.

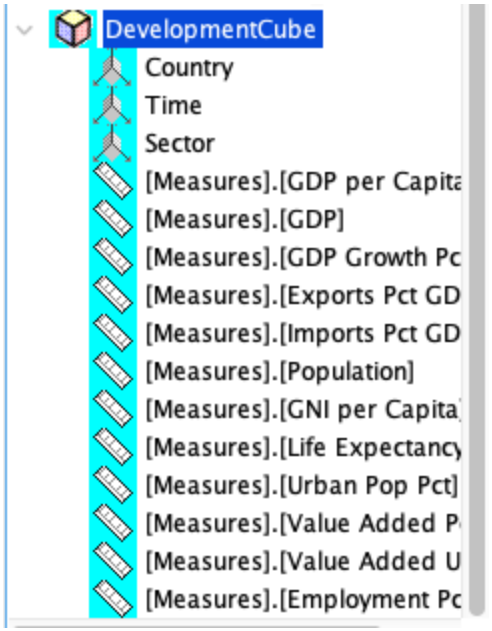
Task 5 – Cube Definition

Cube in Pentaho SW

We developed a cube in Pentaho SW.

- One cube per fact table, since the grains differ: **Economy** (country, year), **Society** (country, year) and **Sectors**(country, year, sector).
- Three shared dimensions, defined once and reused with `DimensionUsage`: **Country** (Country Group → Country), **Time** (Decade → Year) and **Sector** (Broad Sector → Sector).
- Measures are fact-table columns aggregated with `avg`, so higher levels (decade, group, broad sector) show averages.
- A **virtual cube** (`DevelopmentCube`) combines all 3 dimensions and the 12 measures, so Saiku can analyse GDP, society and sector indicators together. Sector only applies to the sector measures.





Analysis A: GDP per capita, per decade and country

Decade	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010	2020
Country	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita
Czechia	-	-	-	-	-	-	-	10,614.50	12,902.30	14,339.53	21,593.44	28,432.68	31,355.13
Hungary	2,681.00	3,266.00	3,459.00	3,942.40	3,653.43	4,713.30	7,002.20	9,108.00	10,596.60	10,318.85	17,179.03	22,940.77	27,876.43
Poland	2,700.00	2,733.00	3,187.00	2,829.56	3,534.90	4,396.80	5,984.50	8,733.70	8,954.10	9,435.85	15,719.68	24,288.02	30,577.73
Romania	1,425.12	1,492.45	1,375.48	1,536.51	1,593.38	2,400.09	3,750.91	5,808.27	6,536.59	5,788.94	11,429.35	19,657.04	25,700.07
Portugal	1,997.10	1,949.70	2,297.80	2,679.00	2,976.80	3,868.60	6,236.20	10,740.10	13,892.30	19,552.67	24,399.78	25,583.82	27,276.97
Estonia	-	-	-	-	-	-	-	13,799.00	16,621.50	14,252.12	21,191.05	24,511.51	29,609.40
Latvia	-	-	-	-	-	-	-	12,506.00	14,466.50	10,470.31	16,110.74	21,717.33	26,470.67
Lithuania	-	-	-	-	-	-	-	12,103.00	13,742.10	10,090.46	15,715.33	23,756.19	30,130.70
Belgium	6,123.80	6,099.80	7,359.10	7,730.90	7,150.70	9,718.70	13,362.70	19,775.90	24,106.70	29,416.07	35,898.08	39,099.71	40,535.53
France	4,666.30	4,996.50	6,305.00	6,891.40	5,661.70	9,731.70	14,367.10	20,827.90	25,021.90	29,539.23	35,167.25	37,296.85	37,740.53
Germany	4,920.50	4,995.80	5,478.90	6,703.80	7,031.10	8,794.20	14,244.10	19,553.70	23,980.30	28,655.99	36,611.38	44,371.89	45,934.40
Greece	2,395.60	2,226.10	3,263.70	3,907.40	2,530.00	3,949.70	6,895.70	12,261.10	14,777.20	17,488.67	25,532.69	23,416.19	23,594.43
Ireland	-	4,361.00	4,185.89	4,704.70	4,950.60	6,028.40	8,061.60	11,501.20	14,865.00	24,832.72	46,398.79	52,404.38	57,556.80
Italy	3,533.60	3,856.20	4,314.10	4,699.80	4,295.60	7,270.40	12,198.70	17,514.10	22,593.20	28,395.32	34,529.31	34,360.77	34,478.90
Spain	2,775.70	2,982.36	3,631.33	3,385.33	3,419.56	4,407.10	7,164.89	12,042.65	15,235.05	21,688.80	30,340.42	32,369.63	32,503.63

Analysis A: GDP per capita, per year and country

Year	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987
Country	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita	GDP per Capita
Czechia	10,954.00	11,317.00	11,634.00	12,055.00	12,164.00	12,336.00	12,306.00	12,109.00	12,301.00	12,573.00	12,855.00	13,211.00	13,514.00
Hungary	9,253.00	9,231.00	9,765.00	9,967.00	9,964.00	10,052.00	10,123.00	10,493.00	10,401.00	10,696.00	10,452.00	10,678.00	10,861.00
Poland	9,258.00	9,397.00	9,483.00	9,741.00	9,471.00	9,149.00	8,584.00	8,429.00	8,764.00	9,006.00	9,022.00	9,240.00	9,059.00
Romania	5,995.50	6,243.10	6,322.20	6,476.20	6,612.00	6,591.60	6,514.10	6,490.40	6,418.20	6,659.50	6,628.60	6,718.40	6,551.90
Portugal	10,388.00	10,861.00	11,422.00	11,700.00	12,326.00	12,822.00	12,934.00	13,198.00	13,158.00	12,894.00	13,240.00	13,774.00	14,641.00
Estonia	-	-	-	-	-	15,044.00	15,478.00	15,927.00	16,361.00	16,660.00	16,541.00	16,866.00	16,898.00
Latvia	-	-	-	-	-	13,739.00	13,970.00	14,190.00	14,376.00	14,682.00	14,273.00	14,539.00	14,360.00
Lithuania	-	-	-	-	-	12,734.00	13,190.00	13,654.00	13,718.00	13,796.00	13,287.00	13,667.00	13,831.00
Belgium	19,831.00	20,916.00	21,025.00	21,605.00	22,094.00	23,060.00	22,760.00	23,071.00	23,071.00	23,644.00	23,873.00	24,221.00	24,772.00
France	20,653.00	21,465.00	22,177.00	22,698.00	23,326.00	23,537.00	23,655.00	24,120.00	24,300.00	24,519.00	24,755.00	25,238.00	25,756.00
Germany	19,193.00	20,218.00	20,837.00	21,447.00	22,305.00	22,497.00	22,553.00	22,380.00	22,840.00	23,564.00	24,133.00	24,657.00	25,027.00
Greece	12,309.00	12,919.00	13,158.00	13,860.00	14,193.00	14,300.00	14,180.00	14,153.00	14,132.00	14,453.00	14,850.00	15,047.00	14,944.00
Ireland	11,662.00	11,639.00	12,425.00	13,150.00	13,335.00	13,614.00	13,893.00	14,061.00	13,931.00	14,435.00	14,834.00	14,768.00	15,458.00
Italy	17,123.00	18,147.00	18,599.00	19,230.00	20,275.00	20,959.00	21,041.00	21,123.00	21,345.00	21,868.00	22,469.00	23,106.00	23,824.00
Spain	12,613.30	13,045.00	13,335.20	13,568.70	13,678.10	14,005.90	13,895.30	14,046.10	14,240.90	14,313.30	14,697.00	15,243.00	16,273.10

Analysis B:
GDP, GNI, Life Expectancy: Portugal vs Western Europe since 1975

Country Group	Focus			Western Europe		
Decade	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy
1980	137,303,463,776	3,707.00	72.78	797,951,956,887	9,813.86	74.96
1990	195,160,217,976	10,585.00	74.94	1,017,259,105,484	20,173.29	77.14
2000	253,695,212,470	16,976.00	77.71	1,336,321,468,888	30,478.29	79.50
2010	264,206,196,129	21,531.00	80.83	1,515,215,326,435	38,819.86	81.78
2020	278,539,477,029	25,830.00	81.80	1,570,620,580,508	46,035.48	82.16

Analysis B:
GDP, GNI, Life Expectancy: Portugal vs Western Europe since 1975

Country Group	Focus			Western Europe		
Year	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy
1975	97,762,401,296	2,210.00	68.30	623,894,674,504	4,954.29	72.47
1976	104,504,213,200	2,330.00	68.90	655,665,251,422	5,244.29	72.73
1977	110,366,212,738	2,430.00	70.00	676,111,757,805	5,637.14	73.10
1978	113,475,729,199	2,530.00	70.30	697,420,232,081	6,447.14	73.29
1979	119,864,921,263	2,870.00	71.20	724,309,257,087	7,994.29	73.63
1980	125,370,949,095	3,350.00	71.20	739,724,211,380	9,628.57	73.77
1981	127,400,923,099	3,420.00	71.60	743,161,406,219	9,554.29	74.10
1982	130,127,665,855	3,230.00	72.40	748,444,251,289	8,620.00	74.43
1983	129,898,975,552	2,830.00	72.30	758,739,888,415	7,600.00	74.50
1984	127,462,176,351	2,520.00	72.50	775,641,322,533	7,354.29	74.96
1985	131,038,827,148	2,530.00	72.80	792,776,619,204	7,357.14	75.07
1986	136,464,680,943	3,080.00	73.30	812,800,625,781	8,442.86	75.31
1987	145,169,747,164	4,160.00	73.70	835,055,012,346	10,750.00	75.63
1988	156,041,295,580	5,630.00	73.70	870,009,235,831	13,905.71	75.87
1989	164,059,396,971	6,320.00	74.30	903,166,995,877	14,925.71	75.99
1990	171,225,929,203	7,130.00	74.00	912,062,949,972	16,201.43	76.19
1991	177,606,537,211	8,180.00	74.00	943,652,774,605	17,384.29	76.34
1992	183,951,331,948	9,860.00	74.30	963,776,236,863	19,537.14	76.70
1993	183,378,088,561	10,100.00	74.50	960,829,829,217	19,304.29	76.77

Analysis B: GDP, GNI, Life Expectancy: Portugal vs Western Europe since 1975

Country	Portugal			Belgium			France			Germany			Greece		
Year	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy
1975	97,762,401,296	2,210.00	68.30	194,240,674,926	6,910.00	72.00	1,114,332,615,000	6,650.00	72.90	1,510,149,923,703	6,740.00	71.30	111,353,885,381	3,420.00	73.40
1976	104,504,213,200	2,330.00	68.90	205,206,876,000	7,580.00	72.10	1,162,522,935,000	7,070.00	73.10	1,583,048,234,281	7,150.00	71.70	118,430,933,161	3,680.00	73.70
1977	110,366,212,738	2,430.00	70.00	206,503,340,893	8,220.00	72.80	1,205,940,906,000	7,730.00	73.40	1,628,636,036,117	7,690.00	72.30	122,480,960,308	3,860.00	73.70
1978	113,475,729,199	2,530.00	70.30	212,370,672,719	9,560.00	72.70	1,239,356,196,000	8,740.00	73.60	1,674,283,009,992	8,890.00	72.30	130,699,231,523	4,510.00	74.40
1979	119,864,921,263	2,870.00	71.20	217,343,101,115	11,730.00	73.20	1,279,104,536,000	10,720.00	73.90	1,741,603,152,539	11,090.00	72.60	135,518,423,132	5,540.00	74.60
1980	125,370,949,095	3,350.00	71.20	227,067,203,496	13,980.00	73.20	1,297,124,070,000	12,950.00	74.10	1,761,467,996,906	13,050.00	72.80	137,887,819,824	6,380.00	74.50
1981	127,400,923,099	3,420.00	71.60	224,240,632,890	13,190.00	73.60	1,310,463,345,000	13,040.00	74.30	1,768,196,429,703	12,580.00	73.00	137,962,177,460	6,250.00	74.90
1982	130,127,665,855	3,230.00	72.40	227,394,760,384	11,180.00	73.90	1,343,411,640,000	11,900.00	74.50	1,753,143,244,687	11,150.00	73.30	138,508,058,074	5,800.00	75.20
1983	129,898,975,552	2,830.00	72.30	227,376,691,105	9,430.00	73.90	1,359,074,700,000	10,320.00	74.80	1,784,298,628,750	10,050.00	73.60	139,066,318,621	5,180.00	75.50
1984	127,462,176,351	2,520.00	72.50	233,018,708,582	8,860.00	74.40	1,379,095,674,000	9,710.00	75.00	1,834,585,161,062	9,900.00	74.10	142,899,252,766	5,040.00	75.80
1985	131,038,827,148	2,530.00	72.80	235,344,813,262	8,820.00	74.50	1,398,409,950,000	9,670.00	75.30	1,874,769,842,531	9,860.00	74.20	147,360,306,005	4,920.00	76.10
1986	136,464,680,943	3,080.00	73.30	238,862,653,069	10,170.00	74.70	1,431,625,550,000	11,230.00	75.60	1,916,181,384,234	11,270.00	74.40	149,738,266,153	5,260.00	76.50
1987	145,169,747,164	4,160.00	73.70	244,504,599,238	13,090.00	75.40	1,467,808,684,000	14,350.00	75.80	1,945,055,815,890	14,280.00	74.80	149,058,782,062	6,020.00	76.10
1988	156,041,295,580	5,630.00	73.70	256,045,020,000	17,010.00	75.60	1,532,315,565,000	18,400.00	76.10	2,009,989,460,679	18,670.00	75.00	155,690,542,546	7,450.00	76.90
1989	164,059,396,971	6,320.00	74.30	265,237,140,019	18,030.00	75.60	1,594,471,896,000	19,360.00	76.30	2,075,675,216,851	19,810.00	75.20	161,665,833,838	8,150.00	76.70
1990	171,225,929,203	7,130.00	74.00	273,278,713,941	19,310.00	76.10	1,636,207,672,000	20,470.00	76.60	2,015,547,498,359	21,360.00	75.10	161,708,974,933	8,810.00	76.90
1991	177,606,537,211	8,180.00	74.00	278,686,606,016	20,580.00	76.20	1,654,070,164,992	21,270.00	76.80	2,141,316,417,831	22,950.00	75.30	167,630,714,744	9,670.00	77.10

Analysis B:
GDP, GNI, Life Expectancy: Portugal vs Eastern Europe since 1990

Country Group	Central/Eastern Europe			Focus			Post-Soviet		
Year	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy
1990	172,897,032,727	3,020.00	70.33	171,225,929,203	7,130.00	74.00	40,062,930,431	-	70.00
1991	158,124,442,560	2,910.00	70.43	177,606,537,211	8,180.00	74.00	36,081,535,132	-	69.60
1992	159,294,373,389	2,505.00	70.58	183,951,331,948	9,860.00	74.30	27,318,273,686	-	69.17
1993	164,864,372,242	2,717.50	70.78	183,378,088,561	10,100.00	74.50	23,517,536,774	-	67.83
1994	174,279,544,410	3,065.00	70.93	186,788,072,120	10,450.00	74.90	22,427,708,751	-	66.90
1995	187,921,303,470	3,555.00	71.08	191,728,089,193	11,210.00	75.30	23,150,588,163	3,020.00	67.63
1996	199,403,857,057	3,977.50	71.33	199,215,852,007	12,100.00	75.30	23,984,379,175	3,370.00	69.50
1997	207,965,822,795	4,117.50	71.53	208,848,444,082	12,370.00	75.40	26,057,523,474	2,963.33	70.00
1998	215,594,740,052	4,132.50	71.98	219,709,918,106	12,190.00	75.70	27,510,786,843	3,160.00	69.87
1999	225,423,610,095	4,180.00	72.23	229,149,917,332	12,260.00	76.00	27,337,160,637	3,246.67	70.47
2000	238,627,001,479	4,352.50	72.78	238,756,377,443	12,180.00	76.30	28,979,273,662	3,543.33	70.90
2001	249,161,680,432	4,412.50	73.20	244,348,167,969	11,850.00	76.80	30,646,217,315	3,763.33	70.93
2002	260,702,632,519	4,705.00	73.25	247,190,712,485	11,780.00	77.10	32,473,789,243	4,100.00	71.23
2003	274,656,970,047	5,630.00	73.35	245,847,246,169	13,290.00	77.20	35,188,218,413	4,986.67	71.57
2004	294,913,099,040	7,045.00	73.68	251,279,602,163	16,190.00	77.70	37,412,326,265	6,376.67	71.97
2005	312,103,233,256	8,520.00	73.85	254,196,923,131	18,600.00	78.10	40,541,218,814	8,143.33	71.77
2006	337,927,171,522	9,642.50	74.23	259,159,946,424	19,410.00	78.40	44,116,963,170	9,633.33	71.57
2007	363,611,473,831	11,012.50	74.43	266,663,287,450	20,860.00	78.30	48,155,522,078	11,583.33	71.57
2008	385,609,611,450	13,127.50	74.70	268,239,264,717	22,630.00	78.50	47,217,857,421	13,563.33	72.67
2009	386,158,617,414	13,482.50	75.00	261,270,596,745	22,970.00	78.70	40,055,915,727	13,470.00	73.60
2010	400,709,501,017	13,542.50	75.33	267,277,408,809	23,030.00	79.00	39,847,307,617	13,086.67	74.07

Analysis B: GDP, GNI, Life Expectancy: Portugal vs Eastern Europe since 1990

Country	Portugal			Estonia			Latvia			Lithuania		
Year	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy
1990	171,225,929,203	7,130.00	74.00	27,066,096,904	-	69.50	42,100,704,929	-	69.30	51,021,989,460	-	71.20
1991	177,606,537,211	8,180.00	74.00	23,940,083,198	-	69.40	36,517,762,608	-	69.00	47,786,759,590	-	70.40
1992	183,951,331,948	9,860.00	74.30	20,201,098,370	-	68.90	24,594,371,373	-	68.40	37,159,351,315	-	70.20
1993	183,378,088,561	10,100.00	74.50	18,206,161,939	-	67.90	21,610,169,740	-	66.70	30,736,278,642	-	68.90
1994	186,788,072,120	10,450.00	74.90	17,713,297,126	-	66.50	21,908,377,653	-	65.70	27,661,451,474	-	68.50
1995	191,728,089,193	11,210.00	75.30	18,632,898,028	3,020.00	67.50	21,602,325,855	-	66.40	29,216,540,607	-	69.00
1996	199,215,852,007	12,100.00	75.30	19,286,856,884	3,370.00	69.60	21,944,517,346	-	68.80	30,721,763,295	-	70.10
1997	208,848,444,082	12,370.00	75.40	21,171,918,456	3,640.00	69.80	23,733,281,647	2,640.00	69.30	33,267,370,320	2,610.00	70.90
1998	219,709,918,106	12,190.00	75.70	21,698,756,967	3,820.00	69.40	25,089,221,158	2,790.00	69.00	35,744,382,405	2,870.00	71.20
1999	229,149,917,332	12,260.00	76.00	21,126,870,271	3,840.00	70.10	25,549,229,736	2,910.00	69.70	35,335,381,903	2,990.00	71.60
2000	238,756,377,443	12,180.00	76.30	23,525,730,943	4,170.00	70.40	26,727,290,777	3,240.00	70.30	36,684,799,266	3,220.00	72.00
2001	244,348,167,969	11,850.00	76.80	24,618,992,841	4,310.00	70.30	28,242,589,693	3,520.00	70.80	39,077,069,410	3,460.00	71.70
2002	247,190,712,485	11,780.00	77.10	25,689,833,693	4,620.00	70.90	30,014,856,985	3,850.00	71.00	41,716,677,051	3,830.00	71.80
2003	245,847,246,169	13,290.00	77.20	27,160,243,367	5,730.00	71.30	32,298,352,002	4,530.00	71.30	46,106,059,871	4,700.00	72.10
2004	251,279,602,163	16,190.00	77.70	28,392,969,678	7,500.00	71.90	34,720,425,840	5,610.00	72.00	49,123,583,277	6,020.00	72.00
2005	254,196,923,131	18,600.00	78.10	30,566,893,529	9,730.00	72.60	38,143,684,659	7,120.00	71.40	52,913,078,254	7,580.00	71.30
2006	259,159,946,424	19,410.00	78.40	33,162,229,951	11,500.00	72.70	42,361,248,688	8,590.00	70.90	56,827,410,871	8,810.00	71.10
2007	266,663,287,450	20,860.00	78.30	35,115,520,992	13,430.00	72.80	46,228,816,935	10,730.00	71.00	63,122,228,308	10,590.00	70.90
2008	268,239,264,717	22,630.00	78.50	32,659,893,005	15,190.00	73.80	44,217,430,207	12,830.00	72.40	64,776,249,050	12,670.00	71.80
2009	261,270,596,745	22,970.00	78.70	27,402,480,486	14,570.00	74.80	37,590,730,012	13,360.00	73.10	55,174,536,683	12,480.00	72.90
2010	267,277,408,809	23,030.00	79.00	27,586,968,284	14,580.00	75.40	35,881,177,042	12,250.00	73.50	56,073,777,524	12,430.00	73.30

Analysis B: GDP, GNI, Life Expectancy: Portugal vs Eastern Europe since 1990

Country Group	Central/Eastern Europe			Focus			Post-Soviet		
Decade	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy	GDP	GNI per Capita	Life Expectancy
1990	186,576,909,879	3,497.94	71.12	195,160,217,976	10,585.00	74.94	27,744,842,307	3,136.36	69.10
2000	310,347,149,099	8,193.00	73.85	253,695,212,470	16,976.00	77.71	38,478,730,211	7,916.33	71.78
2010	463,770,131,453	14,182.25	76.56	264,206,196,129	21,531.00	80.83	47,455,729,217	16,114.00	75.36
2020	567,546,341,635	20,830.42	76.69	278,539,477,029	25,830.00	81.80	57,159,536,934	24,440.56	76.32

Analysis C: agriculture, industry, manufacturing and services as a share of added pct GDP

In the first image to the right, we see the Value added pct GDP in Portugal per Broad Sector. In the second, it's the same but using the sector hierarchy drill-down.
In the third image, we analyze with the drill-down but per year to evaluate variations in the value.

Broad Sector	Value Added Pct GDP
Primary	2.46
Secondary	17.01
Tertiary	63.70

Broad Sector	Sector	Value Added Pct GDP
Primary	Agriculture	2.46
Secondary	Industry	21.05
	Manufacturing	12.97
Tertiary	Services	63.70

	Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
Broad Sector	Sector	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP	Value Added Pct GDP
Primary	Agriculture	4.80	4.60	3.80	3.50	3.30	3.10	3.00	2.70	2.70	2.60	2.30	2.30	2.00	2.00
Secondary	Industry	24.80	25.40	25.70	25.30	24.90	24.50	24.10	23.50	22.60	22.20	21.40	21.10	21.20	20.40
	Manufacturing	15.90	16.60	16.50	16.00	15.60	15.00	14.70	14.20	13.50	13.10	12.60	12.30	12.20	11.90
Tertiary	Services	58.50	57.80	58.40	58.60	59.00	60.00	60.60	61.20	62.10	62.60	63.00	62.90	63.50	64.80

Task 6 – MDX Queries

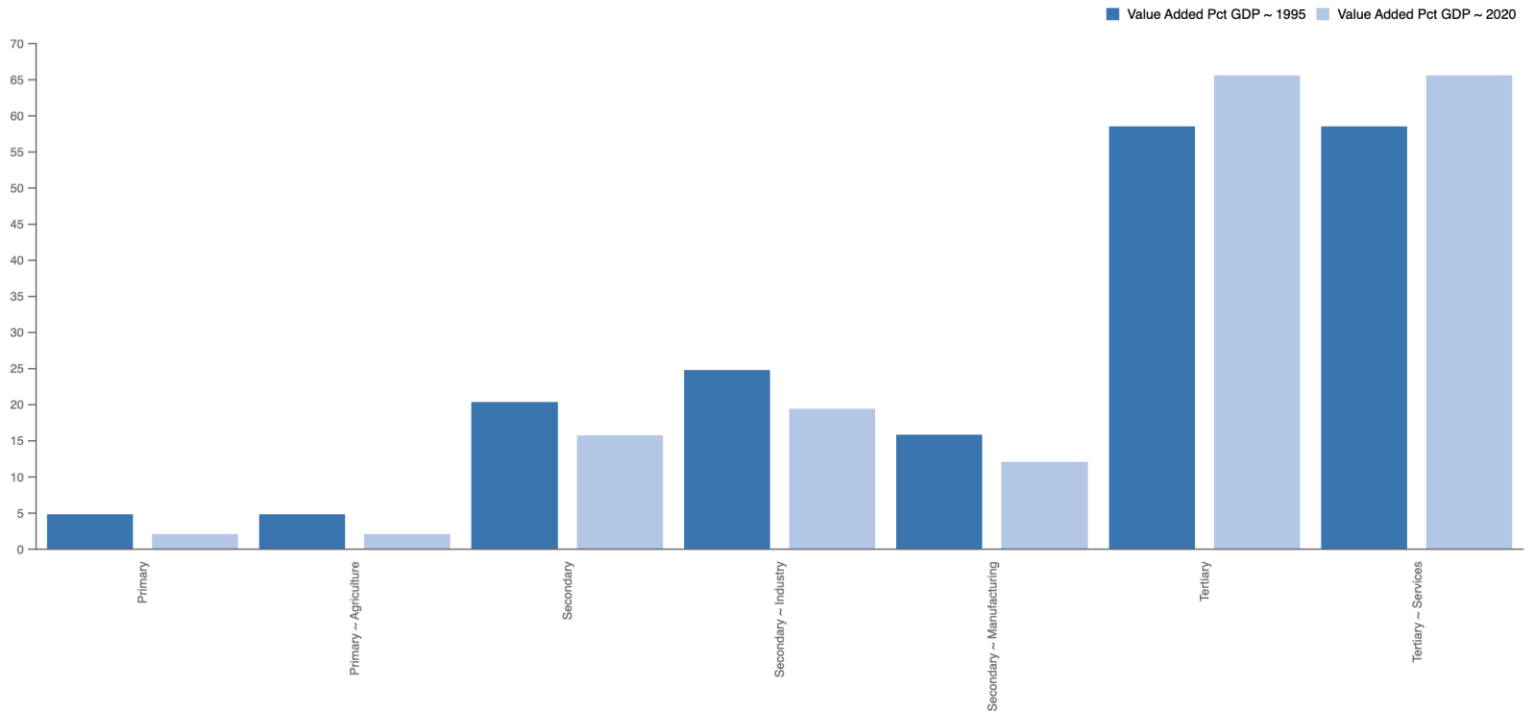
Query 1: Portugal's GDP per capita as a ratio of Spain's GDP per capita from 1960 to 2020

Here we see that Portugal diverged from Spain when it comes to GDP per capita. From 1990 to 2000 a drop of 10% was registered in relation to Spain and since then, the values haven't changed much.

```
WITH
MEMBER [Measures].[GDPpc Portugal] AS
([Measures].[GDP per Capita], [Country].[Focus].[Portugal]),
FORMAT_STRING = '#,##0'
MEMBER [Measures].[GDPpc Spain] AS
([Measures].[GDP per Capita], [Country].[Western
Europe].[Spain]),
FORMAT_STRING = '#,##0'
MEMBER [Measures].[Ratio PT/ES] AS
IIF(IsEmpty([Measures].[GDPpc Spain]) OR [Measures].[GDPpc
Spain] = 0,
NULL,
[Measures].[GDPpc Portugal] / [Measures].[GDPpc Spain]),
FORMAT_STRING = '0.000'
SELECT
{ [Measures].[GDPpc Portugal], [Measures].[GDPpc Spain],
[Measures].[Ratio PT/ES] } ON COLUMNS,
{ [Time].[1960] : [Time].[2020] } ON ROWS
FROM [DevelopmentCube]
```

Decade	GDPpc Portugal	GDPpc Spain	Ratio PT/ES
1960	6,236	7,165	0.870
1970	10,740	12,043	0.892
1980	13,892	15,235	0.912
1990	19,553	21,689	0.902
2000	24,400	30,340	0.804
2010	25,584	32,370	0.790
2020	27,277	32,504	0.839

Query 2: Portugal's value_added_pct_gdp at broad sector and sector level



Broad Sector	Sector	Value Added Pct GDP/1995	Value Added Pct GDP/2020
Primary		4.8	2.1
Primary	Agriculture	4.8	2.1
Secondary	Agriculture	20.35	15.75
Secondary	Industry	24.8	19.4
Secondary	Manufacturing	15.9	12.1
Tertiary	Manufacturing	58.5	65.6
Tertiary	Services	58.5	65.6

Query 2: Portugal's value_added_pct_gdp at broad sector and sector level

In the previous slide we had a graphic image of the query and the table itself.
In order to get those results, we used the following query:

```
SELECT
  { ([Measures].[Value Added Pct GDP],
[Time].[1990].[1995]),
    ([Measures].[Value Added Pct GDP], [Time].[2020].[2020])
} ON COLUMNS,
  Hierarchize(
    Descendants([Sector].[All Sectors], [Sector].[Broad
Sector], SELF_AND_AFTER)
  ) ON ROWS
FROM [DevelopmentCube]
WHERE ([Country].[Country].[Focus].[Portugal])
```

Task 7 – Analytical Questions

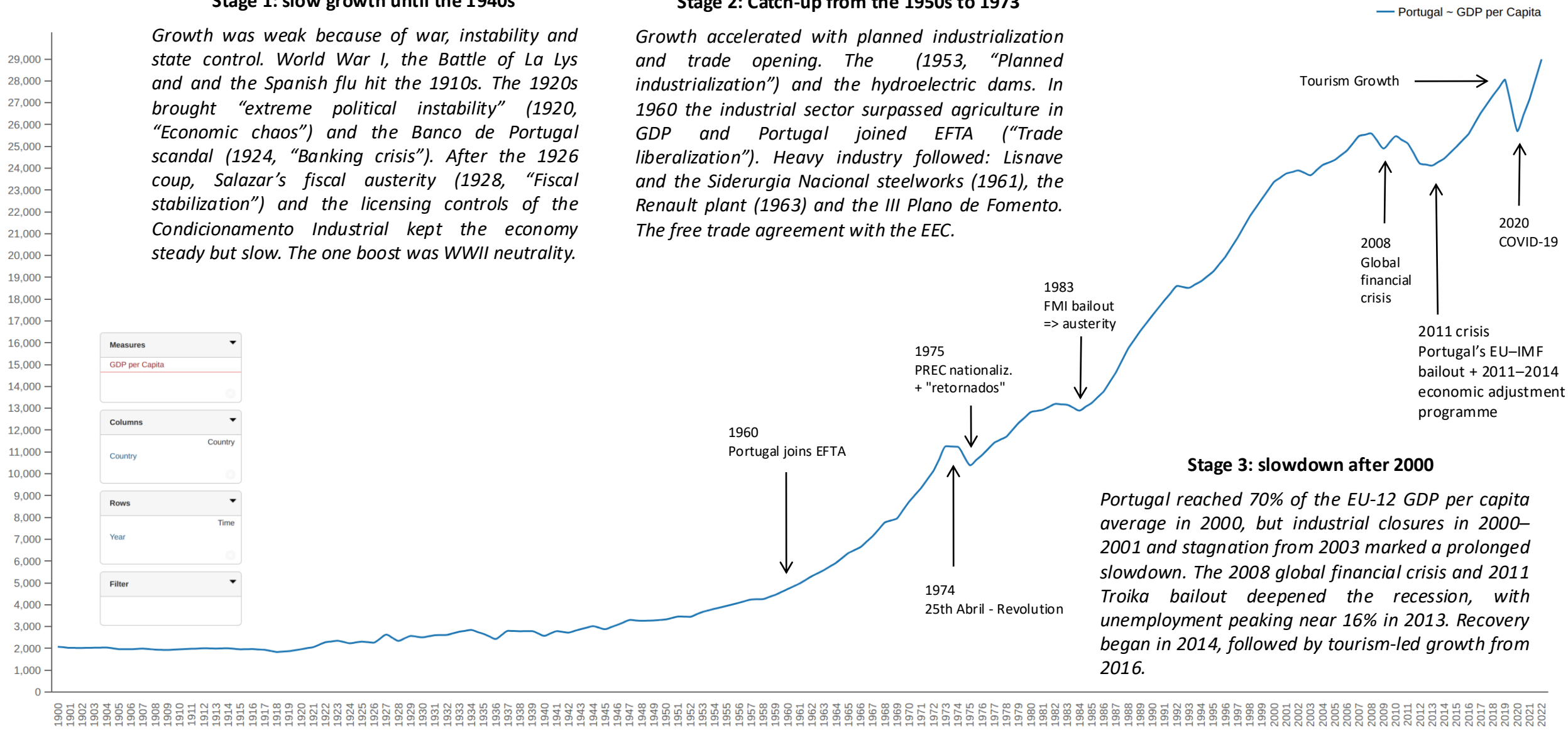
A - Portugal's GDP per capita since 1900

Stage 1: slow growth until the 1940s

Growth was weak because of war, instability and state control. World War I, the Battle of La Lys and and the Spanish flu hit the 1910s. The 1920s brought "extreme political instability" (1920, "Economic chaos") and the Banco de Portugal scandal (1924, "Banking crisis"). After the 1926 coup, Salazar's fiscal austerity (1928, "Fiscal stabilization") and the licensing controls of the Condicionamento Industrial kept the economy steady but slow. The one boost was WWII neutrality.

Stage 2: Catch-up from the 1950s to 1973

Growth accelerated with planned industrialization and trade opening. The (1953, "Planned industrialization") and the hydroelectric dams. In 1960 the industrial sector surpassed agriculture in GDP and Portugal joined EFTA ("Trade liberalization"). Heavy industry followed: Lisnave and the Siderurgia Nacional steelworks (1961), the Renault plant (1963) and the III Plano de Fomento. The free trade agreement with the EEC.



Stage 3: slowdown after 2000

Portugal reached 70% of the EU-12 GDP per capita average in 2000, but industrial closures in 2000–2001 and stagnation from 2003 marked a prolonged slowdown. The 2008 global financial crisis and 2011 Troika bailout deepened the recession, with unemployment peaking near 16% in 2013. Recovery began in 2014, followed by tourism-led growth from 2016.

A - Portugal's GDP per capita since 1900

- Decade growth (%) measures the percentage change in GDP per capita compared with the previous decade.

Decade	GDP per Capita	Decade Growth
1910	1,949.70	-2.4%
2010	25,583.82	4.9%
2020	27,276.97	6.6%
1940	2,976.80	11.1%
1930	2,679.00	16.6%
1920	2,297.80	17.9%
2000	24,399.78	24.8%
1980	13,892.30	29.3%
1950	3,868.60	30.0%
1990	19,552.67	40.7%
1960	6,236.20	61.2%
1970	10,740.10	72.2%
1900	1,997.10	Infinity%

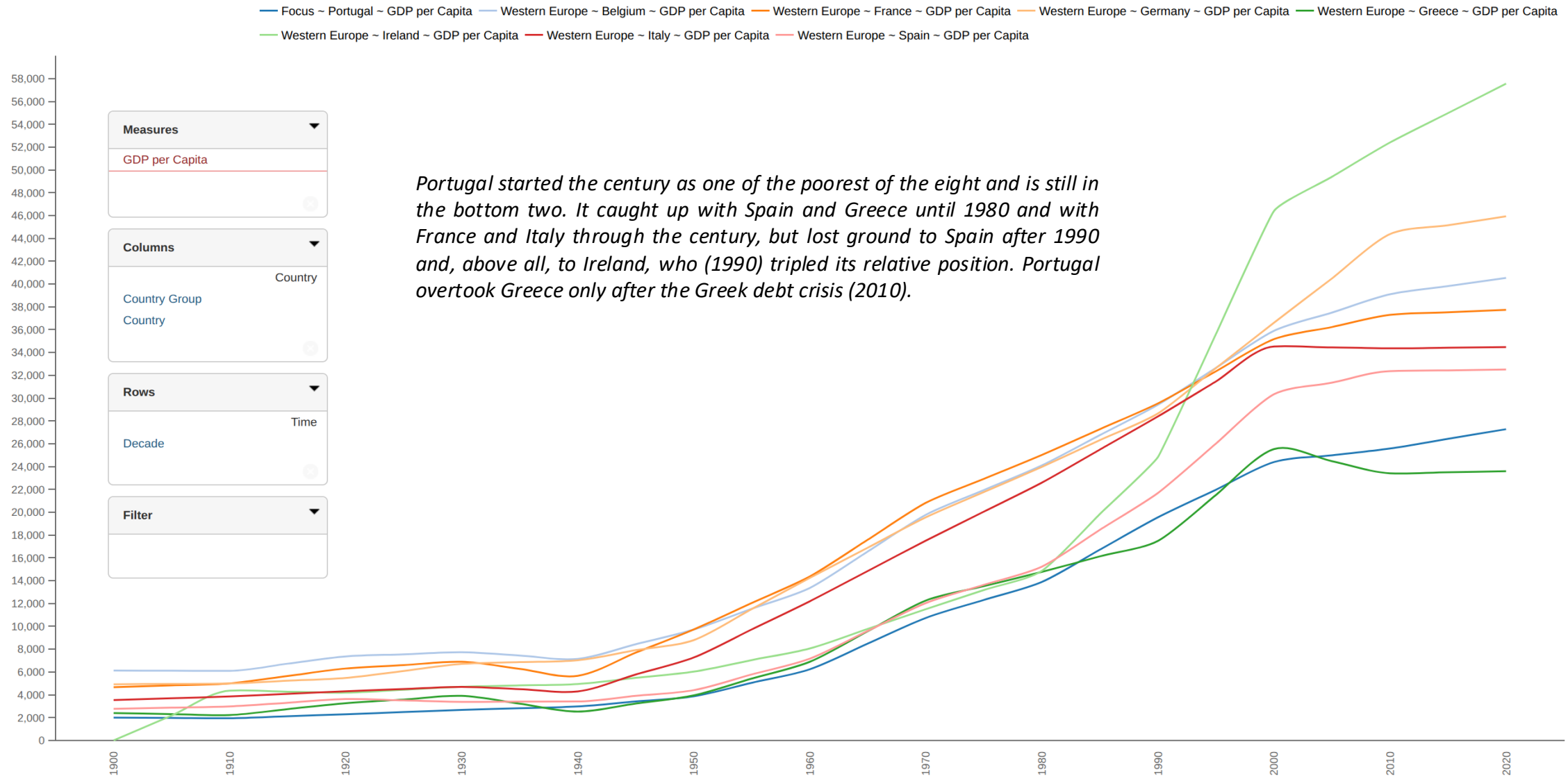
```
WITH
    MEMBER [Measures].[Decade Growth] AS
        ([Measures].[GDP per Capita] /
          ([Measures].[GDP per Capita],
             [Time].[Time].CurrentMember.PrevMember)
        ) - 1,
        FORMAT_STRING = '0.0%'
   SELECT
    {
        [Measures].[GDP per Capita],
        [Measures].[Decade Growth]
    } ON COLUMNS,
    ORDER(
        [Time].[Time].[Decade].Members,
        [Measures].[Decade Growth],
        ASC
    ) ON ROWS
FROM [DevelopmentCube]
WHERE ([Country].[Country].[Focus].[Portugal])
```

Note: A year-level detail of this table can be found in analysis_a.xlsx

- **Fastest decade:** the **1970s** reflects the 1960–1973 boom (EFTA 1960, heavy industry from 1961, free trade with the EEC in 1972) ending with the oil crisis of 1973. Second: the 1960s.
- Third: the 1990s after EEC accession (1986), privatisations and EU-funded infrastructure; the euro came in 1999.
- **Slowest:** the **2010s** with the 2008 crisis and the 2011 Troika bailout; the **1910s** are −2.4% (World War I and the Spanish flu).

- The complete yearly results are available in *analysis_a.xlsx*, provide a long-term view and help identify periods of expansion and contraction. The first-year growth value (1900) is undefined because no previous year is available; the −100% values for 2023–2025 reflect missing GDP-per-capita data.
- Growth accelerated during the 1950s–1970s, reflecting post-war European economic expansion and Portugal's industrialisation, before slowing due to the 1973 oil crisis and the political and economic disruption surrounding the 1974 "Revolução dos Cravos". The sharp contraction in 1975 coincided with this transition.
- Growth was more moderate during much of the 1990s and 2000s, despite Portugal joining the European Economic Community in 1986, and the 2009 decline reflected the global financial crisis. The steep contraction in 2020 (−8.4%) coincided with the COVID-19 pandemic and associated restrictions, followed by a rebound in 2021 (+5.6%) and 2022 (+6.8%) as economic activity recovered.

B/C - Portugal vs. Western Europe countries (Economy - GPD per capita)



Portugal started the century as one of the poorest of the eight and is still in the bottom two. It caught up with Spain and Greece until 1980 and with France and Italy through the century, but lost ground to Spain after 1990 and, above all, to Ireland, who (1990) tripled its relative position. Portugal overtook Greece only after the Greek debt crisis (2010).

B - Portugal vs. Western Europe countries (Economy and Society)

Country Group	Focus		Western Europe	
Decade	Exports Pct GDP	Imports Pct GDP	Exports Pct GDP	Imports Pct GDP
1960	-	-	11.55	14.07
1970	18.32	26.39	23.62	26.21
1980	25.59	33.62	29.62	31.43
1990	26.32	33.94	33.60	32.63
2000	28.51	36.93	42.07	40.94
2010	39.33	40.02	51.88	48.08
2020	44.17	44.67	58.15	52.97

- Portugal and Western Europe experienced an increase in **exports and imports** as a percentage of GDP.
- However, **Portugal consistently recorded an import share above its export share**, whereas Western Europe recorded the reverse pattern from 1990 onward.
- Overall, the results indicate an increasingly balanced relationship between Portugal's exports and imports, while Western Europe maintained higher trade-to-GDP ratios in the most recent reported decades.

Note: A country-level detail of this table can be found in analysis_b.xlsx

Country Group	Focus			Western Europe		
Decade	Urban Pop Pct	Life Expectancy	Population	Urban Pop Pct	Life Expectancy	Population
1900	-	-	5,615	-	-	27,552
1910	-	-	5,976	-	-	28,929
1920	-	-	6,346	-	-	26,201
1930	-	-	7,176	-	-	27,634
1940	-	-	8,004	-	-	28,502
1950	-	-	8,679	-	-	30,232
1960	32.62	65.15	9,084	63.15	70.47	32,708
1970	37.84	68.64	9,320	68.57	72.45	34,958
1980	44.20	72.78	9,881	70.96	74.96	36,108
1990	50.70	74.94	9,975	72.49	77.14	37,286
2000	57.74	77.71	10,395	74.29	79.50	38,701
2010	61.10	80.83	10,331	76.13	81.78	40,181
2020	61.22	81.80	10,212	77.21	82.16	40,761

- **Life expectancy** nearly converged (the gap fell from 5.32 to 0.36 years between 1960 and 2020), after democratisation (1974–76) and EEC accession (1986).
- **Urbanisation** rose from 32.62% to 61.22% but stayed below Western Europe's 77.21%.
- Portugal's **population** peaked in the 2000s and declined after the 2008 crisis and the 2011 bailout, while Western Europe's kept growing.

C - Portugal vs. Western Europe countries (Economy - GPD per capita)

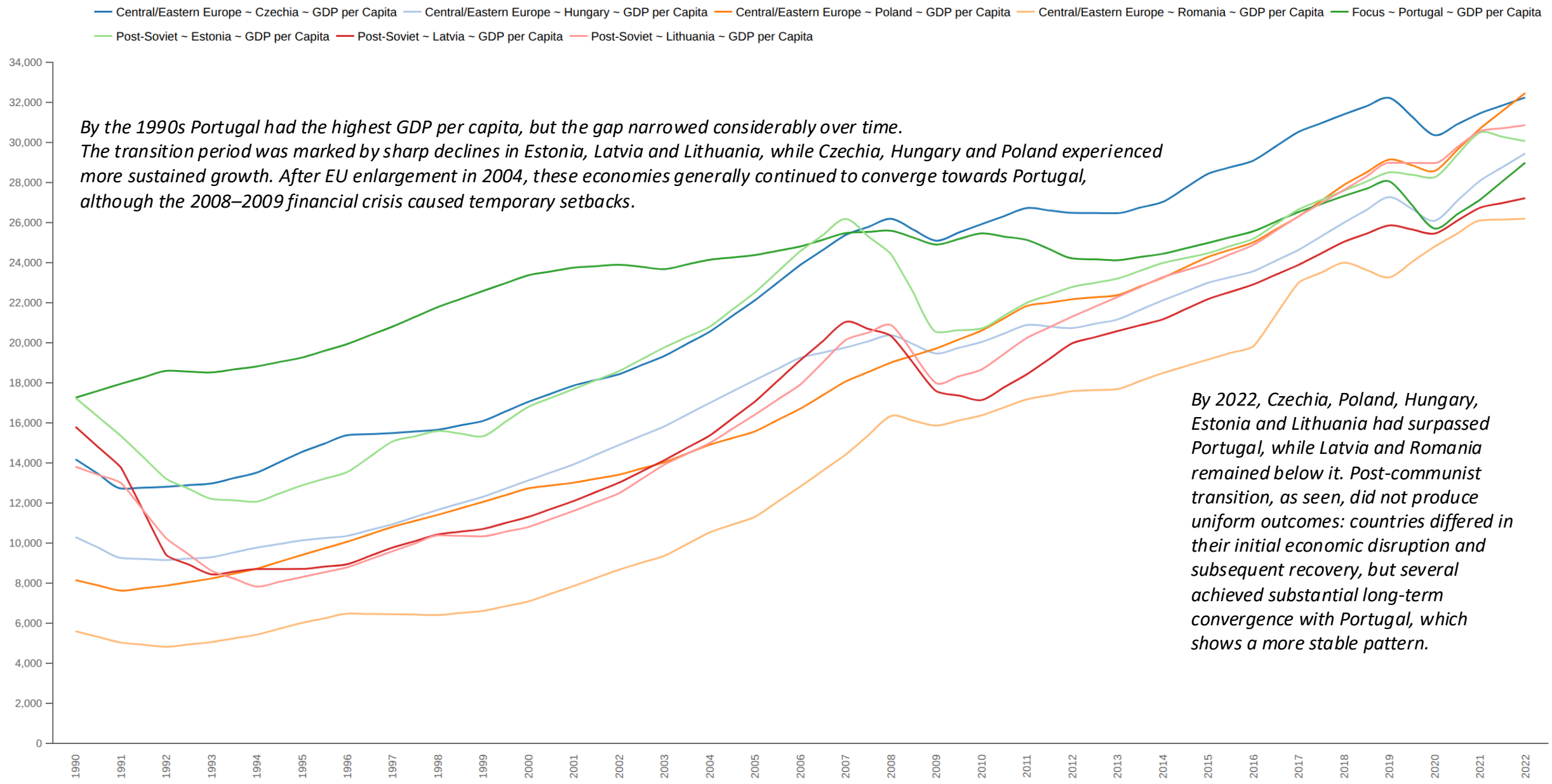
Note: A mdx query to retrieve a country-year-level detail of this table can be found in mdx_c.txt

```
WITH
MEMBER [Measures].[Portugal GDPpc] AS
  (
    [Measures].[GDP per Capita],
    [Country].[Country].[Focus].[Portugal]
  )
MEMBER [Measures].[Western Europe Average GDPpc] AS
  (
    [Country].[Country].[Western Europe],
    [Measures].[GDP per Capita]
  )
MEMBER [Measures].[Portugal Relative GDPpc %] AS
  IIF(
    IsEmpty([Measures].[Western Europe Average GDPpc])
    OR [Measures].[Western Europe Average GDPpc] = 0,
    NULL,
    [Measures].[Portugal GDPpc]
      / [Measures].[Western Europe Average GDPpc]
  ),
  FORMAT_STRING = '0.0%'
SELECT
  {
    [Measures].[Portugal GDPpc],
    [Measures].[Western Europe Average GDPpc],
    [Measures].[Portugal Relative GDPpc %]
  } ON COLUMNS,
  NON EMPTY
    [Time].[Time].[Decade].Members ON ROWS
FROM [DevelopmentCube]
```

Decade	Portugal GDPpc	Western Europe Average GDPpc	Portugal Relative GDPpc %
1900	1,997.10	4,069.25	49.1%
1910	1,949.70	4,195.55	46.5%
1920	2,297.80	4,944.84	46.5%
1930	2,679.00	5,431.90	49.3%
1940	2,976.80	5,005.61	59.5%
1950	3,868.60	7,128.60	54.3%
1960	6,236.20	10,899.26	57.2%
1970	10,740.10	16,210.94	66.3%
1980	13,892.30	20,082.76	69.2%
1990	19,552.67	25,716.69	76.0%
2000	24,399.78	34,925.42	69.9%
2010	25,583.82	37,617.06	68.0%
2020	27,276.97	38,906.32	70.1%

- Portugal's GDP per capita increased substantially but remained below the Western European average in every reported decade.
- Progress was uneven before 1950. Notably, the improvement between 1930 and 1940 reflected both an increase in Portugal's GDP per capita and a decrease in the Western European average.
- Portugal lost relative ground between 1990 and 2010, despite continued increases in its own GDP per capita. Its relative position declined, indicating that the Western European average grew faster over this interval. By 2020, Portugal had recovered to 70.1%, but remained below its 1990 relative position.

D - Portugal vs. Eastern Europe & Post-Soviet countries (Economy)



By the 1990s Portugal had the highest GDP per capita, but the gap narrowed considerably over time.
The transition period was marked by sharp declines in Estonia, Latvia and Lithuania, while Czechia, Hungary and Poland experienced more sustained growth. After EU enlargement in 2004, these economies generally continued to converge towards Portugal, although the 2008–2009 financial crisis caused temporary setbacks.

By 2022, Czechia, Poland, Hungary, Estonia and Lithuania had surpassed Portugal, while Latvia and Romania remained below it. Post-communist transition, as seen, did not produce uniform outcomes: countries differed in their initial economic disruption and subsequent recovery, but several achieved substantial long-term convergence with Portugal, which shows a more stable pattern.

D - Portugal vs. Eastern Europe & Post-Soviet countries (Economy)

- The computation expresses each country's GDP per capita as a proportion of Portugal's GDP per capita in the same decade:

  `Relative GDPpc = Country GDPpc/Portugal GDPpc`

```
WITH
MEMBER [Measures].[GDPpc Relative to Portugal %] AS
  [Measures].[GDP per Capita]
  /
  (
    [Measures].[GDP per Capita],
    [Country].[Country].[Focus].[Portugal]
  ),
  FORMAT_STRING = '0.0%'
SELECT
  {
    [Country].[Country].[Focus].[Portugal],
    [Country].[Country].[Central/Eastern Europe].[Poland],
    [Country].[Country].[Central/Eastern Europe].[Czechia],
    [Country].[Country].[Central/Eastern Europe].[Hungary],
    [Country].[Country].[Central/Eastern Europe].[Romania],
    [Country].[Country].[Post-Soviet].[Estonia],
    [Country].[Country].[Post-Soviet].[Latvia],
    [Country].[Country].[Post-Soviet].[Lithuania]
  } ON COLUMNS,
    Filter(
      [Time].[Time].[Decade].Members,
      [Time].[Time].CurrentMember.Name >= "1990"
    ) ON ROWS
FROM [DevelopmentCube]
WHERE ([Measures].[GDPpc Relative to Portugal %])
```

Decade	Portugal	Poland	Czechia	Hungary	Romania	Estonia	Latvia	Lithuania
1990	100.0%	48.3%	73.3%	52.8%	29.6%	72.9%	53.5%	51.6%
2000	100.0%	64.4%	88.5%	70.4%	46.8%	86.8%	66.0%	64.4%
2010	100.0%	94.9%	111.1%	89.7%	76.8%	95.8%	84.9%	92.9%
2020	100.0%	112.1%	115.0%	102.2%	94.2%	108.6%	97.0%	110.5%

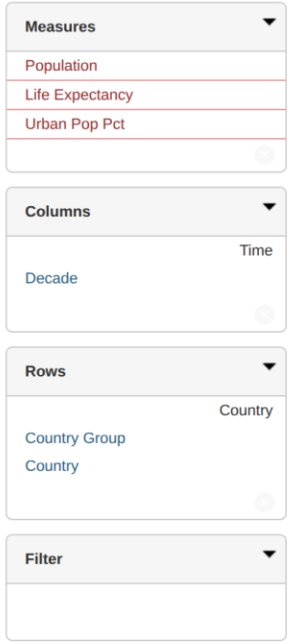
- All seven countries **improved their GDP-per-capita position relative** to Portugal between 1990 and 2020.
- By the 2020s, Poland, Czechia, Hungary, Estonia and Lithuania had GDP per capita **above** Portugal's. Czechia was already above Portugal in the 2010s.
- Romania rose from 29.6% to 94.2% of Portugal's level, the **largest increase in relative terms**. However, it remained below Portugal in the 2020s.
- Latvia reached 97.0% of Portugal's level, coming close but remaining slightly behind.
- The relative positions of these countries improved substantially between the 2000s and 2020s, particularly for Poland, Romania and Lithuania.

D - Portugal vs. Eastern Europe & Post-Soviet countries (Society)

Note: A country-year-level detail of this table can be found in analysis_d.xlsx

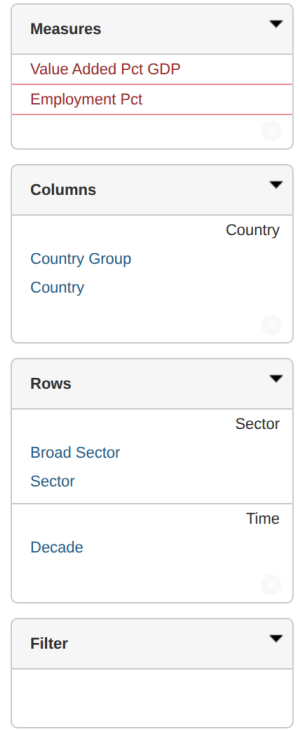
	Decade	1990			2000			2010			2020		
Country Group	Country	Life Expectancy	Urban Pop Pct	Population	Life Expectancy	Urban Pop Pct	Population	Life Expectancy	Urban Pop Pct	Population	Life Expectancy	Urban Pop Pct	Population
Central/Eastern Europe	Czechia	73.12	74.75	10,265	75.95	73.68	10,232	78.52	73.06	10,501	78.82	72.75	10,641
	Hungary	69.85	62.46	10,309	72.71	66.04	10,114	75.55	70.13	9,871	75.74	70.37	9,721
	Poland	71.86	61.54	38,308	74.83	61.50	38,301	77.29	60.37	38,556	77.14	59.92	38,202
	Romania	69.63	53.83	22,678	71.89	53.13	21,536	74.86	53.63	20,069	75.04	52.27	19,371
Focus	Portugal	74.94	50.70	9,975	77.71	57.74	10,395	80.83	61.10	10,331	81.80	61.22	10,212
Post-Soviet	Estonia	68.86	69.28	1,466	72.15	67.52	1,363	77.21	68.52	1,320	78.28	70.45	1,329
	Latvia	68.23	69.38	2,516	71.42	67.92	2,250	74.29	67.90	1,988	74.86	68.35	1,887
	Lithuania	70.20	67.37	3,544	71.76	66.77	3,239	74.57	67.05	2,832	75.82	68.40	2,728

- **Life expectancy** increased in all seven countries, with Estonia recording particularly strong progress, but Portugal remained ahead of every country.
- Portugal's **urban population** share increased from 50.7% to 61.2%, while the comparison countries generally remained more urbanised, particularly Czechia. The only exception is Romania that was caught by Portugal in the 2000s and kept behind.
- Demographic trends differed: Poland and Czechia remained relatively stable, whereas Hungary, Romania and all three Post-Soviet countries experienced **population** declines over the last 2 decades.



D - Portugal vs. Eastern Europe & Post-Soviet countries (Sectors)

	Country Group		Central/Eastern Europe		Focus		Post-Soviet	
Broad Sector	Sector	Decade	Value Added Pct GDP	Employment Pct	Value Added Pct GDP	Employment Pct	Value Added Pct GDP	Employment Pct
Primary	Agriculture	1990	9.74	17.24	4.00	6.76	6.19	18.01
		2000	4.69	12.58	2.47	6.69	3.83	10.84
		2010	3.35	9.00	1.94	4.46	3.25	6.64
		2020	2.79	6.69	2.05	2.93	3.03	4.82
Secondary	Industry	1990	34.20	36.91	25.22	33.92	26.73	30.01
		2000	30.48	34.57	22.09	32.32	25.45	29.47
		2010	29.99	33.08	19.27	25.02	23.58	26.39
		2020	26.95	32.67	18.80	24.80	21.82	25.55
	Manufacturing	1990	21.88	-	16.12	-	16.23	-
		2000	19.58	-	13.06	-	14.56	-
		2010	19.72	-	11.88	-	14.04	-
		2020	16.77	-	12.02	-	12.61	-
Tertiary	Services	1990	46.30	45.87	58.46	59.34	55.65	51.98
		2000	53.54	52.86	62.75	60.98	59.85	59.69
		2010	54.71	57.92	65.87	70.56	61.16	66.98
		2020	59.30	60.63	66.02	72.30	63.31	69.62



- As a common pattern, **agriculture declined**, industry lost relative importance, and **services expanded** across.
- Portugal was already more service-oriented in 1990 and remained so in 2020, when services accounted for 66.0% of GDP and 72.3% of employment. Central/Eastern Europe retained a stronger industrial base, with industry representing 27.0% of GDP in 2020, compared with 18.8% in Portugal. The Post-Soviet group followed an intermediate pattern, with services reaching 63.3% of GDP.
- Agriculture's **employment** share fell particularly sharply in the Post-Soviet group, from 18.0% to 4.8%.

Task 8 – Final Report and Data Quality Discussion

A – Report using PRD (settings)

Objective and scope

- The report compares economic development and living standards across European countries using **GDP per capita** and **life expectancy**, complemented by historical **events**.
- The reporting period is restricted to 1960–2022 to improve temporal overlap between the two indicators.
- Countries are organised by `country_group`, excluding the Benchmark group. Data was retrieved from the warehouse aggerated (average) by `decade`.

Data retrieval

- A JDBC data source connects Pentaho Report Designer (PRD) to the `DevelopmentDW` warehouse.
- The SQL query (on the right) joins country, time, economic and social tables to retrieve country names, groups, decades, GDP per capita and life expectancy.
- Annual observations are aggregated by country and decade using `AVG()`, producing one record per country-decade.
- A `LEFT JOIN` retrieves life expectancy where available while preserving matching economic records when social data are missing.
- Historical events are aggregated separately by country and decade, with descriptions shortened to limit the text length. They are then joined to the main query, adding context without duplicating economic or social observations.

```
WITH events_by_decade AS (
    SELECT
        be.country_key, t.decade, LEFT(
    GROUP_CONCAT(
        CONCAT(t.year, ': ', LEFT(be.event, 50))
        ORDER BY t.year
        SEPARATOR '. '), 200
) AS events
    FROM DevelopmentDW.BRIDGE_EVENT be
    JOIN DevelopmentDW.DIM_TIME t
        ON t.time_key = be.time_key
    WHERE t.year BETWEEN 1960 AND 2022
    GROUP BY be.country_key, t.decade)
SELECT
    c.country_group, c.country_name, t.decade,
    AVG(fe.gdp_per_capita) AS avg_gdp_per_capita,
    AVG(fs.life_expectancy) AS avg_life_expectancy, e.events
FROM DevelopmentDW.DIM_COUNTRY c
JOIN DevelopmentDW.FACT_ECONOMY fe
    ON fe.country_key = c.country_key
JOIN DevelopmentDW.DIM_TIME t
    ON t.time_key = fe.time_key
LEFT JOIN DevelopmentDW.FACT_SOCIETY fs
    ON fs.country_key = c.country_key
    AND fs.time_key = t.time_key
LEFT JOIN events_by_decade e
    ON e.country_key = c.country_key
    AND e.decade = t.decade
WHERE t.year BETWEEN 1960 AND 2022
    AND c.country_group <> 'Benchmark'
GROUP BY
    c.country_group, c.country_name, t.decade, e.events
ORDER BY
    t.decade, c.country_group, c.country_name;
```

A – Report using PRD (structure)

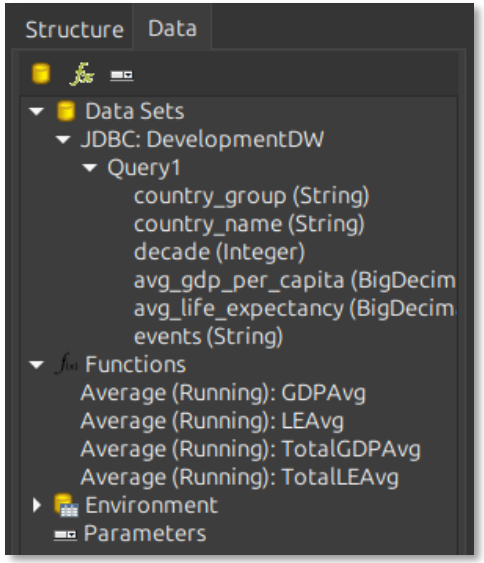
Grouping and layout

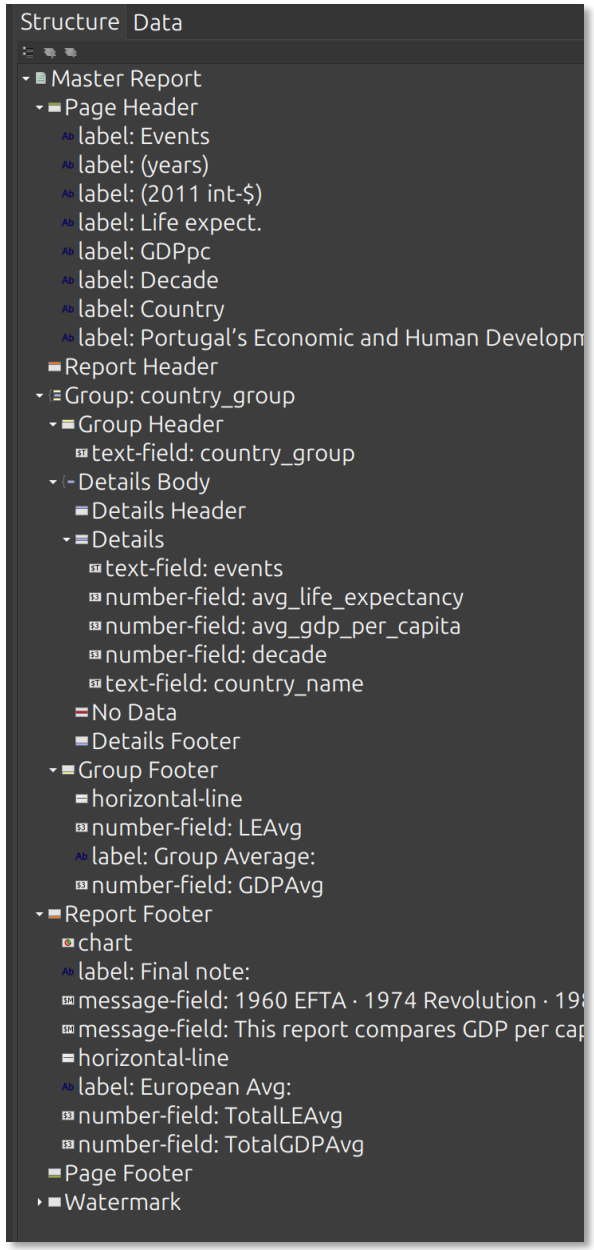
- The report is organised by country group, with countries and decades presented in a grouped table.
- A chart complements the table by illustrating GDP-per-capita trends over time.
- The table provides country-level detail for both chosen indicators, while the chart facilitates visual comparison across countries and periods, regarding, solely, GPD per capita.

Calculated fields

- Four **PRD functions** were defined: two to calculate indicator averages at country-group level (`GDPAvg` and `LEAvg`) and two to calculate averages across the complete report dataset (`TotalGDPAvg` and `TotalLEAvg`).
- In the Data Panel, all functions were created as Average. However, the configuration differs on "Reset on Group Name" field. On `GDPAvg` and `LEAvg` that field was set as `coutry_group`, while on the remaining functions it was left empty.
- Afterwards, a `Number Field` was inserted for each function result. `GDPAvg` and `LEAvg` were placed on Group Footer, while the remaining ones on the Report Footer. In the attributes section of each Number Field, a formula was written `=AVERAGE([function_name])`

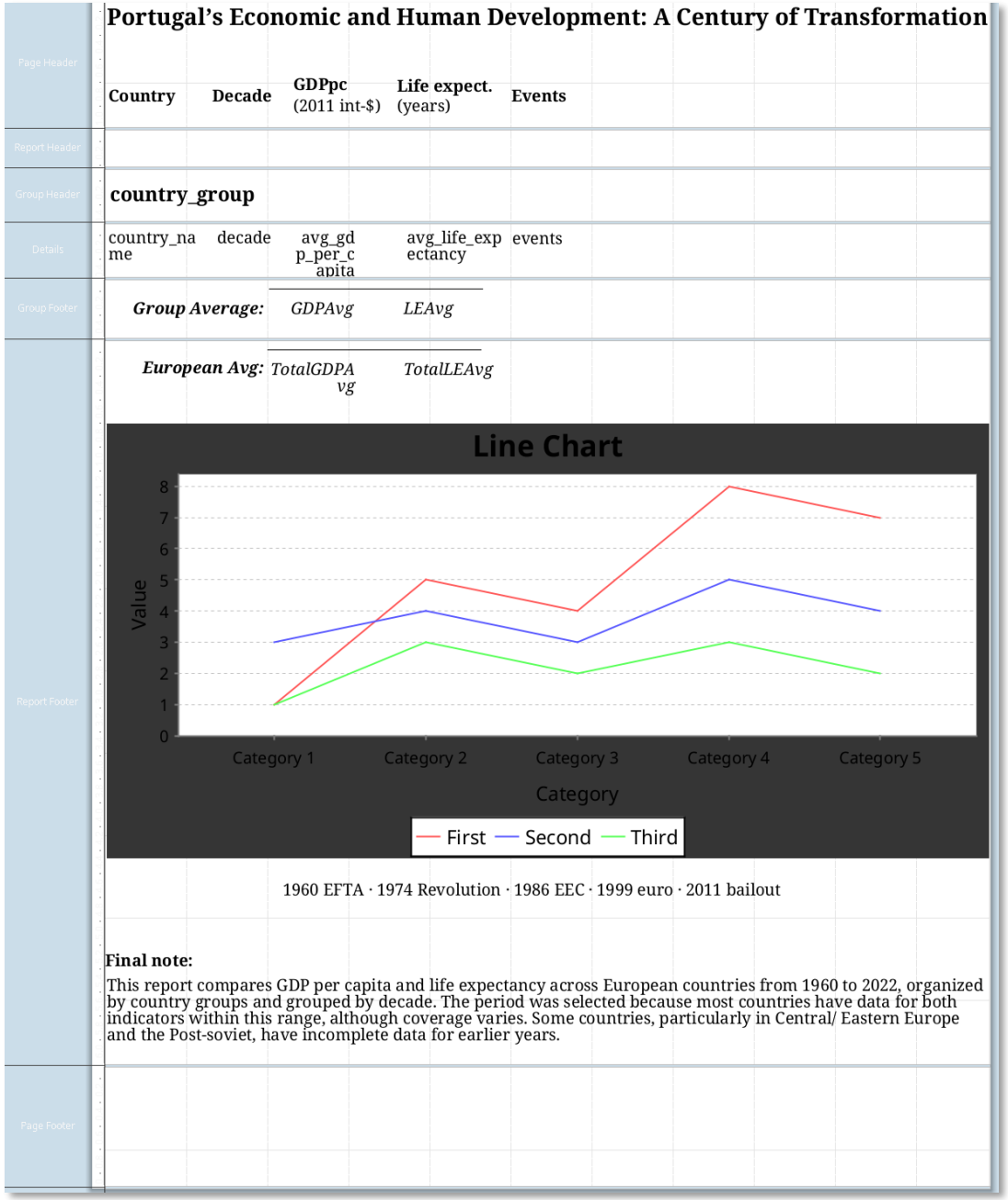
Note: *Since the SQL query already aggregates observations by decade, the report-level averages operate on these decade-level results, not on the original annual records!*





A – Report using PRD (layout)

- The PRD layout is organised into :
  - Page Header band with report title and table headers, both are label elements;
  - Detail band with fields from Data tab (retrieved by the defined query): `country_name`, `decade`, `avg_gdp_per_capita`, `avg_life_expectancy` and `events` (this band is repeated once for every row returned by the query);
  - Group Header with Text Field to `country_group` (this label will appear once at the start of each country group block);
  - Group footer with subtotal averages;
  - Report footer with total averages and chart;



A – Report using PRD (preview)

- The table presents average GDP per capita and life expectancy by country and decade, organised by country group.
- Historical events provide context for selected periods, while group footers display average indicator values across the included decades.

Portugals Economic and Human Development: A Century of Transformation

Country	Decade	GDPpc (2011 int-$)	Life expect. (years)	Events
Central/Eastern Europe				
Hungary	1960	7,002	69.0	1968: New Economic Mechanism; limited market reforms ins
Poland	1960	5,985	68.8	
Romania	1960	3,751	67.0	1965: Ceauescu becomes leader; industrialization and la
Czechia	1970	10,615	70.2	1977: Charter 77; dissident manifesto signed by Havel an
Hungary	1970	9,108	69.4	
Poland	1970	8,734	70.5	1970: Baltic coast strikes; workers protest food prices;
Romania	1970	5,808	69.1	1977: Vrancea earthquake devastates Bucharest; large rec
Czechia	1980	12,902	71.0	1989: Velvet Revolution (November); peaceful end of comm
Hungary	1980	10,597	69.3	1989: Round-table talks; Republic of Hungary proclaimed

Report page 1 (partial)

France	2020	37,741	82.2	2020: COVID-19; France imposes strict lockdowns; GDP con
Germany	2020	45,934	80.8	2020: COVID-19 pandemic; GDP contracts ~4.9%
Greece	2020	23,594	80.7	2020: COVID-19; tourism collapse; GDP contracts sharply
Ireland	2020	57,557	82.4	2020: COVID-19; GDP holds up better than GNI because of
Italy	2020	34,479	82.5	2020: COVID-19; first major European outbreak; GDP contr

Portugals Economic and Human Development: A Century of Transformation

Country	Decade	GDPpc (2011 int-$)	Life expect. (years)	Events
Spain	2020	32,504	82.8	2020: COVID-19; Spain among hardest-hit EU economies; GD
Group Average:		*26,337*	*76.9*	
European Avg:		*21,308*	*74.7*	

Report page 3,4 (partial)

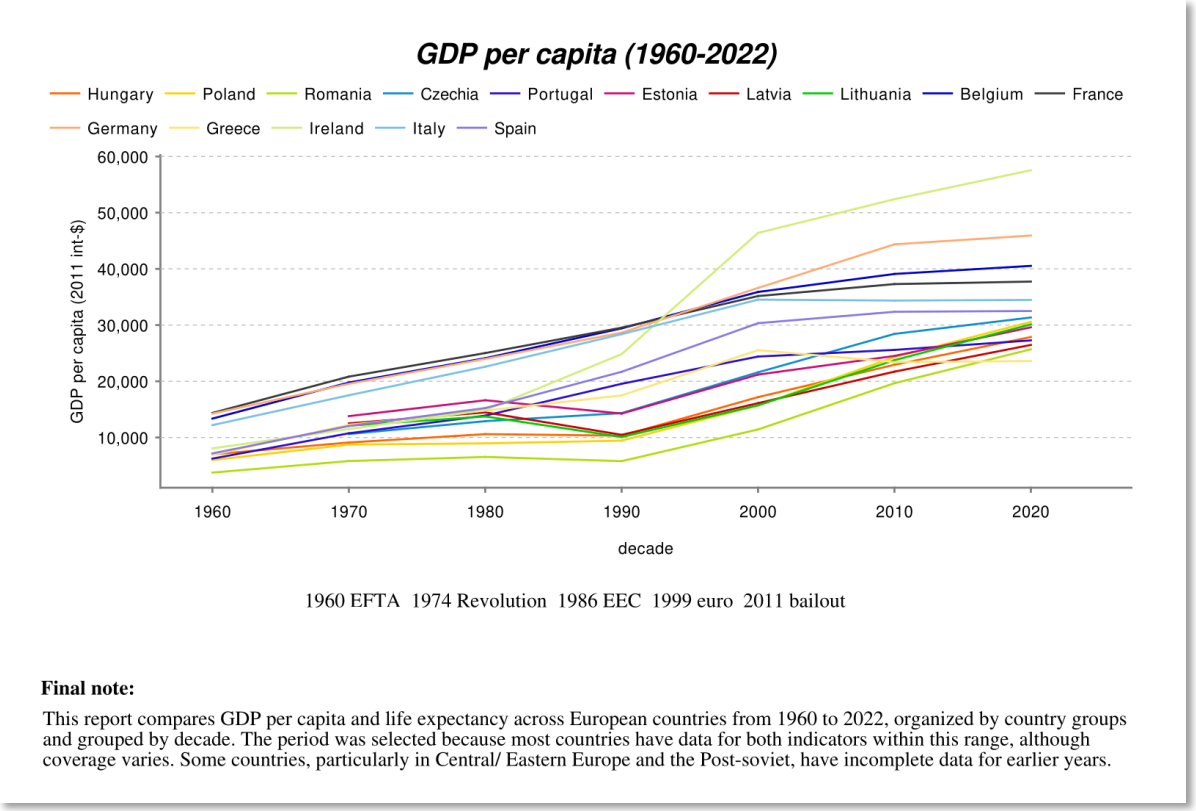
Estonia	2010	24,512	77.2	2011: Estonia adopts the Euro (1 January); first former
Latvia	2010	21,717	74.3	2014: Latvia adopts the Euro (1 January)
Lithuania	2010	23,756	74.6	2015: Lithuania adopts the Euro (1 January)
Estonia	2020	29,609	77.8	2020: COVID-19; GDP contracts ~3%; digital infrastructur
Latvia	2020	26,471	74.2	2020: COVID-19; GDP contracts then rebounds
Lithuania	2020	30,131	74.9	2020: COVID-19; relatively mild GDP contraction
Group Average:		*18,181*	*72.1*	
Western Europe				
Belgium	1960	13,363	70.5	1960: Independence of the Belgian Congo; colonial econom

Report page 2 (partial)

A – Report using PRD (preview)

- The chart illustrates GDP per capita over time, highlighting **long-term economic growth** and differences in development trajectories. Some countries have data only from 1970 onwards, so their series begin later because of differences in temporal data coverage.
- Historical milestones, including the 1974 Revolution, euro adoption in 1999–2002 and the 2011 bailout, help contextualise important periods in Portugal's economic history, corresponding to changes in the slope of the GDP per capita curve.

- It is important to emphasize, once again, that the chart uses decade averages, with each point representing the average GDP per capita for a decade. The lines connect points ten years apart, so annual fluctuations and short-term events within each decade are **not** visible.
- Furthermore, as the chart covers only 1960–2022, the beginning of the 20th century is excluded. However, as observed earlier in the assignment, Portugal's economic growth was relatively slow (aprox. stagnated) in the first half of the century, with stronger growth beginning in the 1960s, during the period of industrialisation and economic expansion (joining EFTA), that preceded the 1973 oil crisis. The major boost, was the euro adopting in 1999, with euro banknotes and coins entering circulation in 2002.



Final note:

This report compares GDP per capita and life expectancy across European countries from 1960 to 2022, organized by country groups and grouped by decade. The period was selected because most countries have data for both indicators within this range, although coverage varies. Some countries, particularly in Central/ Eastern Europe and the Post-soviet, have incomplete data for earlier years.

Report page 4 (partial)

*Despite these limitations, the chart remains **valid** and **meaningful** for a high-level analysis, as decadal averages reveal long-term trends and differences in economic development between countries. Although short-term fluctuations and the beginning of the century are not captured, the analysis supports general conclusions about Portugal's economic growth and its convergence with other European economies.*

B - Data quality

- **Coverage and names.** Maddison has 15 countries; WDI and the events have 16, because the European Union aggregate (EUU) exists only there. Country names and ISO3 codes were consistent across sources, requiring no recoding. However, indicator definitions differ: Maddison measures are explicitly identified, whereas WDI indicator codes encode information such as the measure and sector. WDI codes therefore required interpretation and, where necessary, parsing during ETL to map indicators correctly to the warehouse structure.
- **Time coverage.** Historical events span the entire period ranging from 1900-2025. Maddison spans 1900-2022. WDI starts around 1960, with later starts for specific series: growth and trade from 1990–91 for Czechia, Romania, Poland and the Baltics; sector value added from 1960 for France, 1990 for Italy and Romania, 1991 for Germany and EUU, 1993 for Czechia and 1995 for the rest; employment shares from 1991. No indicator for manufacturing employment exists, so that measure is always empty.
- **Gaps.** All gaps are kept as NULL, never filled. Maddison is unbalanced: Czechia starts in 1970, Estonia, Latvia and Lithuania in 1973, and Ireland in 1913. Seven countries also have internal gaps, the largest being Poland (34 missing years per indicator) and Hungary (23). The measures are not always available together: Romania has GDP per capita for 1901–1919 but no GDP or population for those 17 years. GDP growth contains valid negative values (−32.1% to 42.4%) that are contractions, not errors.
- **Events.** Portugal holds 125 of the 387 events, so the annotation is Portugal-centred (aligned with project purpose) and 165 events (1900–1959) precede WDI and can only be compared with Maddison.
- **Long-to-wide pivot.** WDI stores one row per country, indicator and year, while the facts need one column per measure. The pivot needed input sorted on the grouping keys, exact indicator codes and decimal-safe types. The sector facts also required splitting each code into a sector and a measure.
- **ETL bottlenecks.** One event (Ireland, 1990) had an unquoted comma that shifted its columns while the row count stayed at 387; a content check (16 distinct categories instead of 15) exposed it. With Insert/Update, unsorted input overwrites values silently, so row counts were verified at each step. Insert/Update also writes only the listed fields, so the surrogate keys had to be included in its update grid.
- **Main patterns.** Portugal's GDP per capita rose from 54% (1950) to 76% (1990) of the Western European average, then fell to 68% (2010). EEC accession (1986) precedes the gains, while deindustrialisation, stagnation from 2003 and the 2011 bailout coincide with the reversal. After 1990, five of the seven post-communist countries overtook Portugal.

B - Analytical findings

Most interesting patterns

- **Portugal lost ground relative to Western Europe in GDP per capita after 1990.** Its relative level increased from 54% in 1950 to 76% in 1990, then fell to 68% in 2010 and recovered only to 70% in 2020. The strongest convergence periods were 1963–1973 and 1986–1992. The Western European benchmark includes Ireland, whose unusually high GDP per capita, partly associated with multinational activity, affects the comparison.
- **Several transition economies overtook Portugal.** In 1990, GDP per capita ranged from 32% of the benchmark in Romania to 100% in Estonia. By 2020, five of the seven transition economies had surpassed Portugal. This indicates that Portugal's relative economic position weakened compared with both Western Europe and several former communist economies.
- **Social indicators followed a different trajectory.** Portugal's life-expectancy gap relative to Western Europe narrowed from 5.32 years in 1960 to 0.36 years in 2020. However, the urbanisation gap persisted: 61.22% in Portugal compared with 77.21% in Western Europe. The results therefore differ by development indicator; progress in life expectancy did not imply equivalent convergence in income or urbanisation.
- **Historical events are easier to associate with economic fluctuations than with social or sectoral changes.** GDP growth and other economic indicators can show changes around an event, whereas life expectancy, employment structure and sectoral composition may respond over longer periods. These relationships are temporal associations, not proof of causation. Missing observations and incomplete historical coverage further limit the strength of such comparisons.

! *Overall, the analysis identifies three practical limitations: incomplete overlap between sources, uneven indicator coverage and the risk of silent ETL errors. Despite these constraints, the integrated warehouse supports long-term comparisons and shows that Portugal's relative economic performance, social indicators and structural development did not follow the same trajectory.*

Portugal's Economic and Human Development: A Century of Transformation

Data Analysis and Integration · 2026/2027

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21st October 2026